



Orbital Dynamics in Dwarf Planet Systems

BEng Project Final Dissertation 2023-24

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Date of report: 2nd May 2024

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Summary

This report outlines a bachelor's individual project focused on exploring the complex orbital dynamics of small bodies and their natural satellites. The objective of this exploration was to contribute valuable insights to the field of planetary science and inform future space missions. Specifically, this project conducted a comprehensive study on periodic orbits around Lagrange points for the Restricted Three Body Problem as these orbits offer distinct advantages in the ways of fuel consumption and unobstructed observation points in space.

The project began with a comprehensive literature review followed by the gathering of large amounts of planetary data from various sources. To obtain the periodic orbits, an algorithm was developed in MATLAB that solves the equations of motion for the Circular Restricted Three Body Problem (CRTBP) with a predictor-corrector algorithm and continuation methodology. Once the code had been validated using pre-existing data, Planar Lyapunov and Vertical Lyapunov orbits were obtained for the selected dwarf

planet systems. Application of bifurcation theory in order to obtain more complex families such as the Northern and Southern Halo was also explored, the implication of which is that a method for evaluating and categorising orbital stability was established.

Proceeding this, the fidelity of the model was improved through introduction of perturbations, which accounted for factors like body oblateness and solar radiation pressure, into the equations of motion. The results from the improved model were compared to the initial results and concluded with the effects of any differences.

Completion of this project saw obtained data ready to be compiled into a database for commercial access, this data includes the initial conditions, Jacobi constant, time period and stability of each member in the families obtained. Making this commercially available enables use for design of future mission trajectories.

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Nomenclature

Symbol	Quantity	Unit
C	Jacobi Constant	-
v	Velocity	m/s
μ^*	Mass Ratio Parameter	-
Ω^*	Pseudo Potential Function	-

Subscripts/
superscripts

1	Primary Body
2	Secondary Body

Acronyms

Acronym	Meaning
BVP	Boundary Value Problem
CRTBP	Circular Restricted Three Body Problem
ERTBP	Elliptical Restricted Three Body Problem
IAU	International Astronomical Union
IVP	Initial Value Problem
JWST	James Webb Space Telescope
NASA	National Aeronautics and Space Administration
PALC	Pseudo-Arc Length Continuation
RTBP	Restricted Three Body Problem
STM	State Transition Matrix
SVD	Singular Value Decomposition
TBP	Three Body Problem
TNO	Trans-Neptunian Object
ZVC	Zero Velocity Curve

1. Introduction

1.1. Background

A current focus in the field of planetary science is an exploration into the evolution and habitability of small bodies in the inner and outer solar system. There is speculation that these planets were once somewhat akin to Earth and have now undergone transformative processes to their current state; therefore understanding the evolutionary trajectory of dwarf planets offers a unique perspective on the lifespan and habitability of terrestrial planets. [1]

Sample collection missions play a pivotal role in furthering this understanding, by analysing samples we can obtain results about their chemical and isotopic compositions which can be extrapolated to draw conclusions about the parent body evolution and fundamental processes occurring in the solar system. [2] [3] By extending this sample collection to exoplanets we enrich the database of planetary data thus providing a gateway for deciphering factors that govern universal habitability. From this, a critical research gap emerges – the need for a nuanced understanding of the dynamical environment surrounding dwarf planets which this project addresses by delving into the orbital intricacies, through employing an increasing-fidelity approach.

This project also holds relevancy in the practical application of informing deep space missions, which are inherently linked into the planetary science advancement described above. The contemporary scientific landscape is marked by ambitious exploratory missions. Most prevalently the New Horizons mission has provided unprecedented insight on dwarf planets, capturing the first detailed images of Pluto, and now venturing into the Kuiper Belt wherein numerous (un)discovered dwarf planets reside. [4][5] There are also interplanetary endeavours, notably NASA's Dawn spacecraft which was the first to orbit two extra-terrestrial destinations [6]. Furthering our understanding of small body orbital dynamics leads to a more informed and strategic approach to deep space exploration.

The particular focus of this project is on the periodic orbits around Lagrange points in small body systems. The Restricted Three Body Problem (RTBP) describes the motion of three bodies under their mutual gravitational attraction, therefore this can be used to model the movement of a satellite in a planet-moon system. Lagrange points are equilibrium solutions to the RTBP and are of interest as at a Lagrange point the gravitational fields of the two larger bodies (i.e. the planet and its moon) in the RTBP reach an equilibrium. At this location, a third smaller body (i.e. the satellite) can maintain

a relatively stable orbit for minimal energy expenditure, which is not only desirable but a fact being exploited already for the Earth-Moon and Sun-Earth systems. Most recently, the positioning of the James Webb Space Telescope (JWST) at L_2 for the Sun-Earth system. [7] Located beyond Earth's orbit in the direction opposite the Sun, L_2 provides an unobstructed vantage point for space observation, ensuring continuous monitoring of cosmic events by minimising interference from Earth's shadow. Additionally, the Sun-Earth L_1 point, located between the Earth and the Sun, offers distinct advantages. Positioned there, spacecraft can continuously monitor the solar wind, providing critical information about space weather and potential solar storms that may impact Earth's magnetic field and technological infrastructure. This location also allows for early detection and warnings of approaching solar storms, facilitating preventive measures to protect satellite communication systems, power grids, and astronauts in space missions. [8][9] Thus, extending this understanding to the nature of Lagrange points in the dwarf planet systems opens new doors in the way of deep space exploration and furthering our understanding in the field of planetary science. To further emphasise the widespread application of Lagrange point orbits, Table 1 displays a summary of current and past missions stationed at these points.

Table 1: Table listing all current and past missions to Lagrange points, highlighting the significance of their exploitation and thus the relevancy of this study [10].

Mission	Lagrange Point
International Sun-Earth Explorer 3	Sun-Earth L_1
Advanced Composition Explorer	Sun-Earth L_1
Deep Space Climate Observatory	Sun-Earth L_1
LISA Pathfinder	Sun-Earth L_1
Solar and Heliospheric Observatory	Sun-Earth L_1
WIND	Sun-Earth L_1, L_2
Wilkinson Microwave Anisotropy Probe	Sun-Earth L_2
Herschel Space Telescope	Sun-Earth L_2
Planck Space Observatory	Sun-Earth L_2
Chang'e 2	Sun-Earth L_2
Gaia Space Observatory	Sun-Earth L_2
Spektr-RG	Sun-Earth L_2
JWST	Sun-Earth L_2
Euclid	Sun-Earth L_2
Aditya-L1	Sun-Earth L_1

ARTEMIS	Earth-Moon L_1 , L_2
Chang'e 5-T1 Service Module	Earth-Moon L_2
Queqiao	Earth-Moon L_2

1.2.Aim and Objectives

This research project aims to develop and validate numerical methods for studying orbit families in dwarf planet systems, enhancing our understanding of the dynamical environment, and informing practical applications in deep space exploration.

The objectives can be summarised as follows:

- To obtain periodic orbits using numerical methods in the following dwarf planet systems:
 - o Pluto/Charon; Pluto/Styx; Pluto/Nix; Pluto/Kerberos; Pluto/Hydra
 - o Eris/Dysnomia
 - o Haumea/Hi'iaka; Haumea/Namaka
- Progress the dynamical model through levels of fidelity, starting from low (Classic Restricted Three Body Problem) to mid/high fidelity (perturbed CRTBP).
- Compare and validate model results with other models, existing research, and commercially available software.

2. Literature Review

2.1.The Three Body Problem and Its Derivatives

The foundation for the Three Body Problem (TBP) was laid by Sir Isaac Newton (1643 – 1727) in 'Principia', Leonhard Euler (1707-1783) then introduced the RTBP by considering the third mass to be infinitesimally small; Euler also introduced the rotating reference frame. Joseph-Louis Lagrange (1736-1813) found the equilibrium solutions for the Circular Restricted Three Body Problem (CRTBP) and as such these were named 'Lagrange points'.

The TBP is the motion of three bodies under the influence of their mutual gravitational attractions and does not have a solution in closed form, however it is possible to make assumptions to reduce problem complexity and obtain practical solutions. The most

common starting point is the CRTBP which assumes distance between the two primaries is constant.

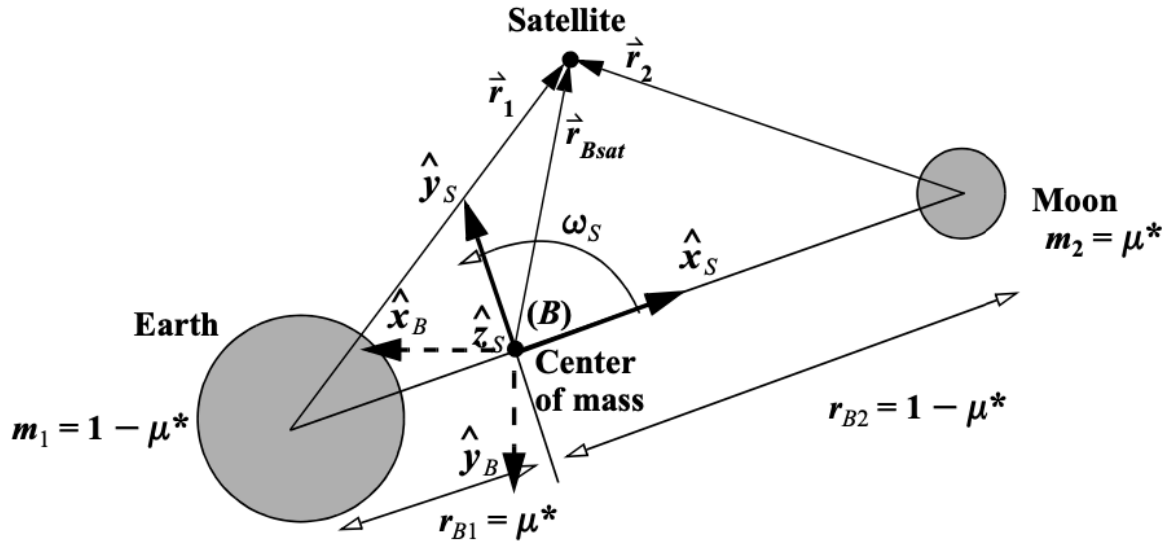


Figure 1: Visual representation of the dimensionless geometry of the CRTBP, specifically for the Earth-Moon system thus in the Earth-Moon rotating frame of reference [11]

The CRTBP considers two primaries (planetary bodies in this case), reduced to point masses, orbiting in circular motion around the common barycentre, the mass of the third body (satellite) is assumed negligible. Satellite motion is described in a rotating coordinate system relative to the primary body m_1 (where $m_1 > m_2$). In the non-dimensional problem, unit of mass is the sum of the primaries and the unit of distance is the separation of these two bodies, angular velocity can therefore be normalised to equal 1 and the mass parameter μ^* (representing the mass ratio of the secondary body to the entire system) can be defined. This is all shown in Figure 1.

In rotating coordinates, the x-axis is the vector between the primaries, the z-axis is parallel to the angular velocity of m_1 and y completes the orthogonal axes. The third body's state vector is represented by $\mathbf{X} = \{x \ y \ z \ \dot{x} \ \dot{y} \ \dot{z}\}$ and the equations of motion for CRTBP can be written as:

$$\begin{aligned} \ddot{x} - 2\dot{y} - x &= -\frac{(1-\mu^*)(x+\mu^*)}{r_1^3} - \frac{\mu^*(x-1+\mu^*)}{r_2^3} \\ \ddot{y} + 2\dot{x} - y &= -\frac{(1-\mu^*)y}{r_1^3} - \frac{\mu^*y}{r_2^3} \\ \ddot{z} &= -\frac{(1-\mu^*)z}{r_1^3} - \frac{\mu^*z}{r_2^3} \end{aligned} \quad (1-3)$$

Where r_1 and r_2 are the vector distances between the third body and m_1 and m_2 , respectively and can be defined using (4) and (5).

$$r_1 = \sqrt{(x + \mu^*)^2 + y^2 + z^2} \quad \text{and} \quad r_2 = \sqrt{(x + \mu^* - 1)^2 + y^2 + z^2} \quad (4) \quad (5)$$

Or, if a pseudo-potential function is defined to be $\Omega^* = \frac{1-\mu}{r_1} + \frac{\mu}{r_2} + \frac{x^2+y^2}{2}$ (6) then equation set (1-3) can be rewritten as

$$\ddot{x} = 2\dot{y} + \Omega_x^*$$

$$\ddot{y} = -2\dot{x} + \Omega_y^* \quad (7-9)$$

$$\ddot{z} = \Omega_z^*$$

Where the subscript of the potential denotes which variable it is partial derivative to.

From the equations of motion, the equilibrium solutions or Lagrange points, where the gravitational forces balance can be obtained for the x-y plane by setting $\ddot{x} = \dot{x} = \ddot{y} = \dot{y} = 0$, there are five total. If the third body is placed in the region of these points, it will maintain its position relative to the other two bodies throughout its orbit, a recent exploitation is positioning JWST at L_2 for the Sun-Earth system; its sunshield is deployed towards the Sun and will remain this way for the mission duration [7].

The first three Lagrange points (L_1 , L_2 and L_3) are co-linear and unstable, lying along the x-axis of the barycentric system (see Figure 2), hence a disturbance would cause an orbiting body to break out of orbit and drift off. Contrastingly, L_4 and L_5 , the triangular points, are stable hence if disturbed, there would be a restoring force and orbiting would continue; L_4 and L_5 lie the same magnitude of distance away from the secondary but are on opposite sides of the x-axis (see Figure 2).

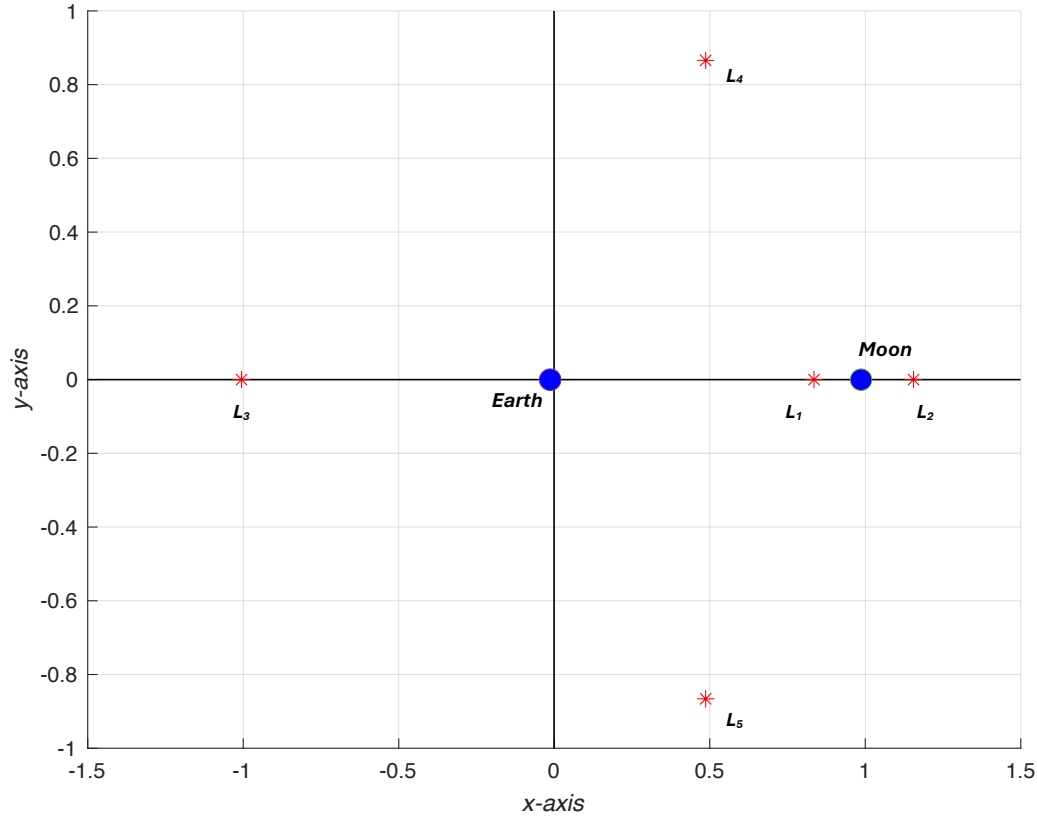


Figure 2: Lagrange point visualisation for the Earth-Moon CRTBP system, $\mu^* = 0.0121505856$, in the Earth-Moon rotating frame of reference

Another key concept for understanding orbital dynamics is the energy pseudo-integral, Jacobi Constant, given by:

$$C = 2\Omega^* - v^2 = x^2 + y^2 + \frac{2(1-\mu^*)}{r_1} + \frac{2\mu^*}{r_2} - v^2 \quad (10)$$

Where x and y are coordinates in the phase space, r_1 and r_2 are the dimensionless relative vectors for the two major bodies, μ^* is the mass parameter, v is velocity of the body moving through the region and Ω^* is the pseudo-potential function.

By setting $v = 0$, ‘forbidden regions’ (or ‘zero velocity curves’, ZVC) can be mapped, with this information, the regions of space accessible through velocity-potential exchange can be identified, which becomes of special interest when trying to calculate escape trajectories and interplanetary transfer gateways [11][12].

2.2.Computing Periodic Orbits

To obtain periodic orbits around Lagrange points, numerical methods can be employed that use the equations of motion for the CRTBP to compute stable trajectories.

Predictor-corrector algorithms work by iteratively refining an initial guess to satisfy constraint criteria, in this context the criteria would be periodicity and phase conditions so that the algorithm converges on periodic orbits. There are various candidates for the predictor-corrector algorithm components hence the subsequent sections of this literature review will only explore the most widely adopted approaches.

2.3. Predictor Algorithms

2.3.1. Differential Correction

Differential correction is the most intuitive approach, both [12] and [14] have used it for obtaining Lagrange point orbits. Considering an initial condition $\bar{x}_0$ and desiring a final position x_d at time t_1 , if the trajectory evolves naturally the final position $\bar{x}_1$, can be described as $\bar{x}_1 = \bar{x}_d - \delta\bar{x}_1$, where $\delta\bar{x}_1$ represents the distance between the desired and actual trajectories at time t_1 . The idea is to reduce this difference to be within a certain tolerance (ε) which is achieved by tweaking the initial conditions. The process is captured by (11) which describes a flow map for a trajectory ending at t_1 , starting at t_0 with initial condition $x_0 + \Delta x_0$ where Δx_0 is the accumulation of corrected $\delta\bar{x}_0$

$$\phi(t_1; t_0; \bar{x}_0 + \Delta\bar{x}_0) = x_d + \varepsilon \quad (11)$$

this turns out to be the commonly known Newton's method where the desired final state is the root of the flow map which is obtained through solving

$$\delta\bar{x}_0 = \Phi(t_1, t_0)^{-1} \delta\bar{x}_1 \quad (12)$$

Where $\Phi(t_1, t_0)$ is the State Transition Matrix (STM), the STM describes the linear relationship between initial and final displacements.

2.3.2. Pseudo-Arc Length Continuation

Pseudo-arc length continuation (PALC) is more robust than Newton's method and is most often employed where trajectories have more complex structures, for example [15] uses this approach when designing a satellite transfer into a Halo orbit. Where differential correction focuses on minimising error between an initial guess and the desired solution, PALC focuses on following a predefined path to reach said solution. Small steps are taken along a path defined by the tangent to the current solution. If u_0 is a solution to $F(u) = 0$ and $\mathbf{u}'$ is the unit tangent vector to the curve at u_0 then the methodology works by solving

$$(u_1 - u_0)^T - \delta s = 0 \quad (13)$$

Where δs is the step size, sufficiently small so that the direction of the branch is not affected. A first guess to u_1 solution is defined by

$$u'_0 = u_0 \pm \delta s u' \quad (14)$$

The guess is then iteratively corrected using (13) as a constraint in Newton's Method.

2.4. Single vs Multiple Shooting Correctors

Correctors refine the orbits obtained by predictor methods by adapting them to satisfy phase and periodicity conditions. Single shooting takes an initial guess at the starting point of the desired interval then integrates the equations of motion over the entire interval using this initial guess. The final state is compared to the desired state and the difference is compared to a tolerance, if below tolerance then the solution satisfies both the initial and final conditions, if not then the initial guess is adjusted based on this error and the process is repeated. This was the approach used by [12] when solving for the Lyapunov family.

Multiple shooting also aims to satisfy periodicity and phase conditions through Newton's Method but breaks the problem into smaller intervals instead of integrating over the entire interval at once. Because of this, it requires initial guesses for the solution at the start of each sub-interval and integrates the equations of motion over each sub-interval, imposing additional constraints to ensure the solutions from each segment 'match up'. In [14] a 'two level correction scheme' has been implemented, this is the same as multiple shooting.

2.5. Bifurcation Theory

Bifurcations are changes in dynamical system parameters that cause a qualitative change in the system dynamics, in this context a bifurcation can affect the stability of a periodic orbit, either creating new families or destruction of existing ones. Family creation at critical bifurcation points can be exploited for obtaining more complex families such as the Halo family; this specific creation/destruction of families happens at the transition between stable and unstable orbits which is a tangent bifurcation [16][17]. This theory introduces the need for a way to analyse orbit stability.

2.6. Orbit Stability

The best established approach for evaluating an orbit's stability is using eigenvalues of the Monodromy matrix, M , which is the STM propagated over one period. All orbits in this study have six eigenvalues existing in reciprocal pairs (any eigenvalue of M , λ_j , has a corresponding partner $1/\lambda_j$). Due to the periodicity condition, two eigenvalues are unity. The remaining nontrivial eigenvalues determine local orbital stability, values of $|\lambda_j| > 1$ correspond to unstable modes and $|\lambda_j| < 1$ correspond to stable modes, for a periodic

orbit to be stable all eigenvalues of M must have a magnitude less than one [12][18]. In CRTBP, eigenvalues of the 6x6 matrix M can be found using (15):

$$\begin{aligned}\det(M - \lambda I_n) &= (\lambda - 1)^2(\lambda - \lambda_1)\left(\lambda - \frac{1}{\lambda_1}\right)(\lambda - \lambda_2)\left(\lambda - \frac{1}{\lambda_2}\right) \\ &= (\lambda - 1)^2(\lambda^2 + b_1\lambda + 1)(\lambda^2 + b_2\lambda + 1)\end{aligned}\quad (15)$$

Where $\lambda_1, \frac{1}{\lambda_1}, \lambda_2, \frac{1}{\lambda_2}$ are the non-trivial, non-unity values of M and $b_1 = -(\lambda_1 + \frac{1}{\lambda_1})$ and $b_2 = -(\lambda_2 + \frac{1}{\lambda_2})$ are stability indices, which according to [28] can be found using

$$b_1, b_2 = \frac{\alpha \pm \sqrt{\alpha^2 - 4\beta + 8}}{2}\quad (16)$$

Where $\alpha = 2 - \text{trace}(M)$ and $\beta = \frac{\alpha^2 - \text{trace}(M^2)}{2} + 1$

These indices can be used for identifying bifurcation points as a stable periodic orbit will have $\max(|\text{Re}(b_1)|, |\text{Re}(b_2)|) < 2$ hence the transition between $b_j \neq 2$ and $b_j = 2$ will correspond to a tangent bifurcation.

3. Methodology

The Methodology section of this report explains the solution reached in order to satisfy the project aim and objectives. The section follows the chronological timeline of the project, hence commences with solution structure and its basis before proceeding on to implementation of steps to achieve said solution. The final portion in the Methodology section is an illustrative example of increasing model fidelity to account for effects on the orbital environment such as body oblateness and axial tilt.

3.1. Data Availability

After conducting the literature review, a preparatory stage that had to happen before construction of the computer model could begin was determination of what systems were to be considered in this study. The classification of whether a body is a dwarf planet or not is actually widely variable across the scientific landscape. Beginning with the opinion of an official authority, the IAU defines a dwarf planet as ‘an object in orbit around the Sun that is large enough to pull itself into a nearly round shape but has not been able to clear its orbit of debris’ which roughly translates to the object having sufficient mass so that it assumes a hydrostatic equilibrium (resulting in a round shape) [19]. By this definition, the IAU only recognises five dwarf planets in our solar system Pluto, Ceres,

Haumea, Makemake and Eris. Unfortunately, current technology is not advanced enough to enable direct measurement of the quantities required to determine whether a body satisfies this definition which introduces the variability in the opinion of what bodies do and what don't. To address this, there have been various attempts to better establish criteria for this definition by trying to use quantitative factors that can be measured directly, like body diameter. The remainder of this section will briefly explain the three most widely adopted criteria after the IAU's then a conclusion of the status of main dwarf planet candidates will be made using each.

3.1.1 Tancredi

Gonzalo Tancredi, a member of the IAU, submitted a report to the IAU of dwarf planet candidates that are currently classified as TNOs (Trans-Neptunian Objects). His proposition was based on light-curve amplitude analysis and the condition the object was more than 450km in diameter. His report proposed 15 dwarf planets by this criteria, which included the five already identified by the IAU and another three in the report, Sedna, Orcus, and Quaoar, had previously been advised to the union to be accepted as dwarf planets [20]. The IAU has not yet responded meaning these latter three still aren't officially recognised.

3.1.2 Brown

Mike Brown, discover of Eris, introduced a spectrum like ordering as opposed to binary categorisation, he sorted bodies by how likely they are to be dwarf planets based on diameter as this usually directly correlates to hydrostatic equilibrium, this approach can be summarised as follows:

- *Nearly certainly*: diameter estimated/measured to be over 900km which almost definitely satisfies the hydrostatic equilibrium condition
- *Highly likely*: diameter estimated/measured to be over 600km, the size would have to be 'grossly in error' or primarily rocky (dwarf planets are assumed to be more icy than rocky) to not be a dwarf planet
- *Likely*: diameter estimated/measured to be over 500km, there are large measurement uncertainties meaning some in this category are likely to be a lot smaller and thus not dwarf planets
- *Probably*: diameter estimated/measured to be over 400km, expected to be dwarf planets if this measurement is correct and they are icy
- *Possibly*: unsure where the transition from non-round to round shape occurs but for icy satellites it is between 200 and 400km, therefore objects in this range may satisfy the criteria (hydrostatic equilibrium) but there is no way to tell for certain
- *Probably not*: no icy satellite below 200km is round.

[21]

Based on this, as of 2020 there were 10 ‘nearly certain’ dwarf planets, five of which are the IAU accepted ones and the remaining are Gonggong, Quaoar, Sedna, Orcus, 2002 MS₄ and Salacia. Sedna, Quaoar, and Orcus align with Tancredi’s findings.

3.1.3 Grundy et al.

Many TNOs within the size range of 400–1000 km exhibit notably low densities around 1.0–1.2 g/cm³, contrasting with denser dwarf planets like Pluto, Eris, and Ceres [22]. Brown suggested that such low-density TNOs are predominantly composed of ice instead of rock, attributing the unexpected low density to the porous nature of ice. Leading on from this, Grundy et al. proposed that TNOs under 1000 km, with low density and albedo, may lack planetary differentiation (the process by which denser material sinks to the core of a forming body resulting in a layered structure), distinguishing them from bodies like Orcus, Quaoar, and Charon [22]. The delineation between these populations appears to be around 900–1000 km, with a suggested upper limit of 600–700 km for significant porosity retention. The result of this is that Pluto, Eris, Haumea, Gonggong, Makemake, Quaoar and Sedna are the only candidates to fulfil Grundy et al.’s criteria with current observational data.

3.1.4 Shortlisting Candidates

Table 2 summarises which of the main dwarf planet candidates are recognised by each of the above bodies

Table 2: Comparison of the results of each criteria against nominal candidates for dwarf planet registration. Y = is a dwarf planet, N = isn't a dwarf planet, M = potentially a dwarf planet and N/A is the study was conducted pre-discovery of the body hence unable to categorise.

	Classified as Dwarf Planet			
	IAU	Tancredi et al.	Brown	Grundy et al.
Pluto	Y	Y	Y	Y
Eris	Y	Y	Y	Y
Haumea	Y	Y	Y	Y
Makemake	Y	Y	Y	Y
Gonggong	N	N/A	Y	Y
Quaoar	Y	Y	Y	Y
Ceres	Y	Y	Y	Y
Orcus	N	Y	Y	Y
Sedna	N	Y	Y	Y
Salacia	N	M	Y	M

Using Table 2 as reference, Pluto, Eris, Haumea, Makemake, Quaoar, Ceres, Orcus, and Sedna were chosen as they each are approved by all or the majority of bodies. Unfortunately, the sparsity of dwarf planet data became an issue again as despite the dwarf planets themselves having calculated/observed masses the mass of their moons is hard to find a reliable source for and both are needed for calculating mass parameter. In order to maintain accuracy in this study, only systems to which sufficient data for reliably determining the mass parameter were used and these were:

- Pluto/Charon, Pluto/Nix, Pluto/Styx, Pluto/Hydra, Pluto/Kerberos
- Eris/Dysnomia
- Haumea/Hi'iaka, Haumea/Namaka

Hence these were the systems shortlisted into the project objectives, Table 3 contains the mass ratio parameter used in the numerical methods for each of these systems.

Table 3: Table of the shortlisted dwarf planet systems used in this model alongside their mass parameters (μ^) [22,23,24,25,26]*

System	Mass Parameter
Pluto-Charon	0.10857221044
Pluto-Nix	0.00000345355
Pluto-Styx	0.000000575594
Pluto-Hydra	0.00000368379
Pluto-Kerberos	0.00000122793
Eris-Dysnomia	0.00488618758
Haumea-Hi'iaka	0.00447984071
Haumea-Namaka	0.00049975012

3.2.Zero Velocity Curves

This project is an investigation into the dynamical environment surrounding dwarf planets by considering the movement of a third body in the vicinity of a (dwarf) planet-moon system, an important concept to understand for accurate interpretation of results obtained by the predictor-corrector is the Jacobi constant and its connection to zero velocity curves. Equation (10), established in the literature review, is the equation for the Jacobi constant and takes the form of $2\Omega^*$ or:

$$C = x^2 + y^2 + \frac{2(1 - \mu^*)}{r_1} + \frac{2\mu^*}{r_2} - v^2$$

This equation reflects the law that total energy in a system is the sum of its potential and kinetic energies, practically this means if a satellite has some velocity v and some potential Ω^* it can travel further by exchanging this kinetic energy for potential.

Considering this exchange, there will be some point where $v = 0$ and $C = \Omega^*$ which means the satellite can travel no further; for a given C value some regions of space will

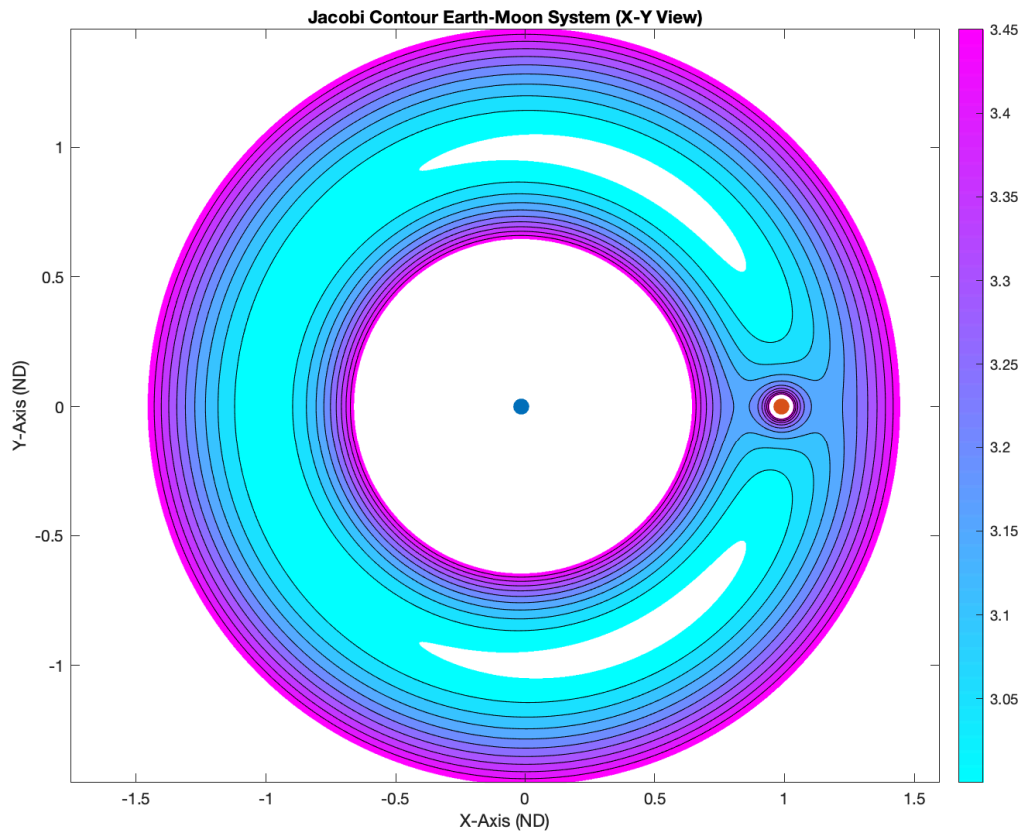


Figure 3: Zero velocity curves for $C = 3.05-3.45$ in the Earth-Moon system. Blue dot is Earth and red dot is the Moon.

remain inaccessible and these are called zero velocity curves. Equation (10) can be solved for various C values to calculate corresponding ZVCs, Figure 3 shows the ZVC for the Earth-Moon system .

The blue dot represents Earth and the red the Moon, the colour bar on the right corresponds to the Jacobi constant magnitude, it should be noted that C is measured negative to the system so a value closer to 0 is actually a larger energy value (-3.05 is closer to 0 than -3.45). For a C value of magnitude 3.45 in this case, if the satellite were to be launched from Earth it would be unable to escape to enter an orbit around the moon or further as it would be unable to cross the magenta ring surrounding Earth. If the C value was changed to 3.05, the inaccessible region would correspond to the 'C' like shape of the cyan contour and so the spacecraft could have a trajectory even to the L_2 point of the Earth-Moon system. At a given point, a spacecraft's velocity and position can be known meaning its corresponding C value can be calculated which is particularly useful for determining whether a spacecraft can execute a free-return trajectory or not. In

summary, although the Jacobi constant doesn't provide any information about initial conditions themselves, it can show the possible trajectories for a given set of initial conditions.

3.3.Solution Concept

Families of periodic orbits in dwarf planet systems are to be obtained through solving the equations of motion for both the unperturbed and perturbed CRTBP with computational methods. The model created will first consider the lower fidelity unperturbed CRTBP primarily for verification and validation purposes, explained further in the next paragraph, before complexity of the model is increased through introduction of perturbations which consider characteristics that influence orbital trajectory, such as body oblateness. Due to the problem's non-dimensional nature, the only variable that has to change when solving for different dwarf planet systems is the mass parameter input, everything else about the code structure can remain the same which is why this was the determining factor in the data collection stage, on which systems were to be considered.

It is intended, to achieve the third objectives laid out in the Aims and Objectives section, that the lower fidelity model will be validated using the publicly available NASA three body periodic orbit database; this database supplies the initial conditions for precomputed Lagrange Point periodic orbit families for the CRTBP alongside a preview of each family for the given systems. Re-iterating here that the defining parameter for a system in the CRTBP is mass ratio μ^* , the same Earth-Moon mass parameter used in the NASA tool can be used in the computational model created for this project and the results compared for each Lagrange Point family. Matching results validates that the algorithm is executing the employed numerical methods correctly, enabling application of the model to the shortlisted dwarf planet systems. It is important to note that the validation is limited to results for the unperturbed Earth-Moon system as the commercial database only has precomputed orbits for the Earth-Moon, Sun-Earth, Sun-Moon, Mars-Phobos, Jupiter-Europa, Saturn-Enceladus, and Saturn-Titan systems, not for any dwarf planet systems. Therefore, once the transition is made to obtaining families for the systems in Table 3 the reliability of these results hinges upon the confidence instilled from the Earth-Moon validation. This reliability also applies to the perturbed problem of the higher fidelity model.

The decision to use the perturbed CRTBP as a higher fidelity model instead of transitioning into the Elliptical Restricted Three Body Problem (ERTBP) or taking the time to create a mesh-based approach over each body, comes at the conclusion of evaluating model flexibility and scope of the project. The perturbed CRTBP allows the model to

remain extremely flexible thus adaptable to each individual system as the definition of the perturbations themselves can vary system to system. For example, Haumea takes on the shape of a triaxial ellipsoid therefore the perturbation that should be considered here is an oblateness coefficient which will account for this shape, but for something like Pluto which is primarily spherical the impact of oblateness will become so small the difference in orbit due to this influence alone will be almost negligible. With regards to project scope, whilst time must be set aside for the sourcing of perturbation equations themselves, implementing them is feasible within the time constraints, doing a second literature review and planning out a reliable validation process before creating the methodology for a new model like ERTBP is not.

3.4. Program Structure

Minimising computational resources and time required for execution creates the most efficient algorithm and after evaluating the potential ways forward, with the top two candidates being presented in the literature review, it was decided that a combination of Pseudo Arc Length Continuation and Single Shooting methods would achieve this. CRTBP involves complex solution spaces due to the highly non-linear nature of the problem, PALC tends to be more robust than Newton's Method for cases where multiple solutions exist and can achieve higher accuracy at less computational cost due to how the equations are solved. Additionally, PALC is especially more efficient when exploring solution curves over a range of parameter values which, for the CRTBP, can correspond to variation of the Jacobi Constant, an important consideration for escape velocity as shown by zero velocity curves. Moreover, casting perspective to the outcomes required for achieving the project aim and objectives, exploiting the bifurcation theory outlined in the literature review will enable the Halo orbit family to be obtained and PALC is particularly well suited for this as its global tracking enables easy identification of bifurcation points.

When compared with multiple shooting, single shooting is conceptually simpler and easier to implement into the algorithm since it involves solving the entire trajectory in one step. The implication of this is fewer iterations and overall a less computationally intense approach, aligning with the goal of minimising computational resources. The main advantage of multiple shooting methodology over single shooting is its handling of trajectories that are complex such as those with multiple revolutions (i.e. quasiperiodic) but considering the project aim, which is specific to periodic orbits, this advantage is less significant. Another reason for choosing single over multiple shooting which only

becomes relevant with the chosen methodology being a predictor-corrector, is that single is easier to incorporate into continuation methods such as PALC.

Once the approach to the program has been decided, its structure must be planned which is commonly presented in the form of flow charts which enable visualisation of the program's logical progression. Table 4 contains the symbols used in flow charts and what operation they represent.

Table 4: Legend for flowchart presented in Figure 4

Symbol	Name	Function
	Start/end	Indicative of a start or end point i.e. program initiation and termination
	Input/Output	User data input or operation results displayed to the user
	Process	Any operation that modifies data or controls the flow of the process i.e. calculations and data manipulation
	Decision	Decision point or conditional statement, binary option with outcomes True/False or Yes/No

After thorough planning, development and optimisation, the flow chart in Figure 4 was the final one that continued forward as the basis for the computer model, presenting the logical progression of the main program body alongside the connections between different stages.

To obtain different families, changes must be made to the initial conditions/step direction for the predictor-corrector however the actual execution of the program stays the same; even using bifurcation theory is a means to obtaining initial conditions for complex families. Therefore, not all potential families that can be obtained with this methodology will be presented in this report, primarily because presenting them all would prove slightly redundant as there is no significant change between the approach used. The results that have been included in this report have been chosen in order to demonstrate application of the literature review theory and the program structure outlined previously.

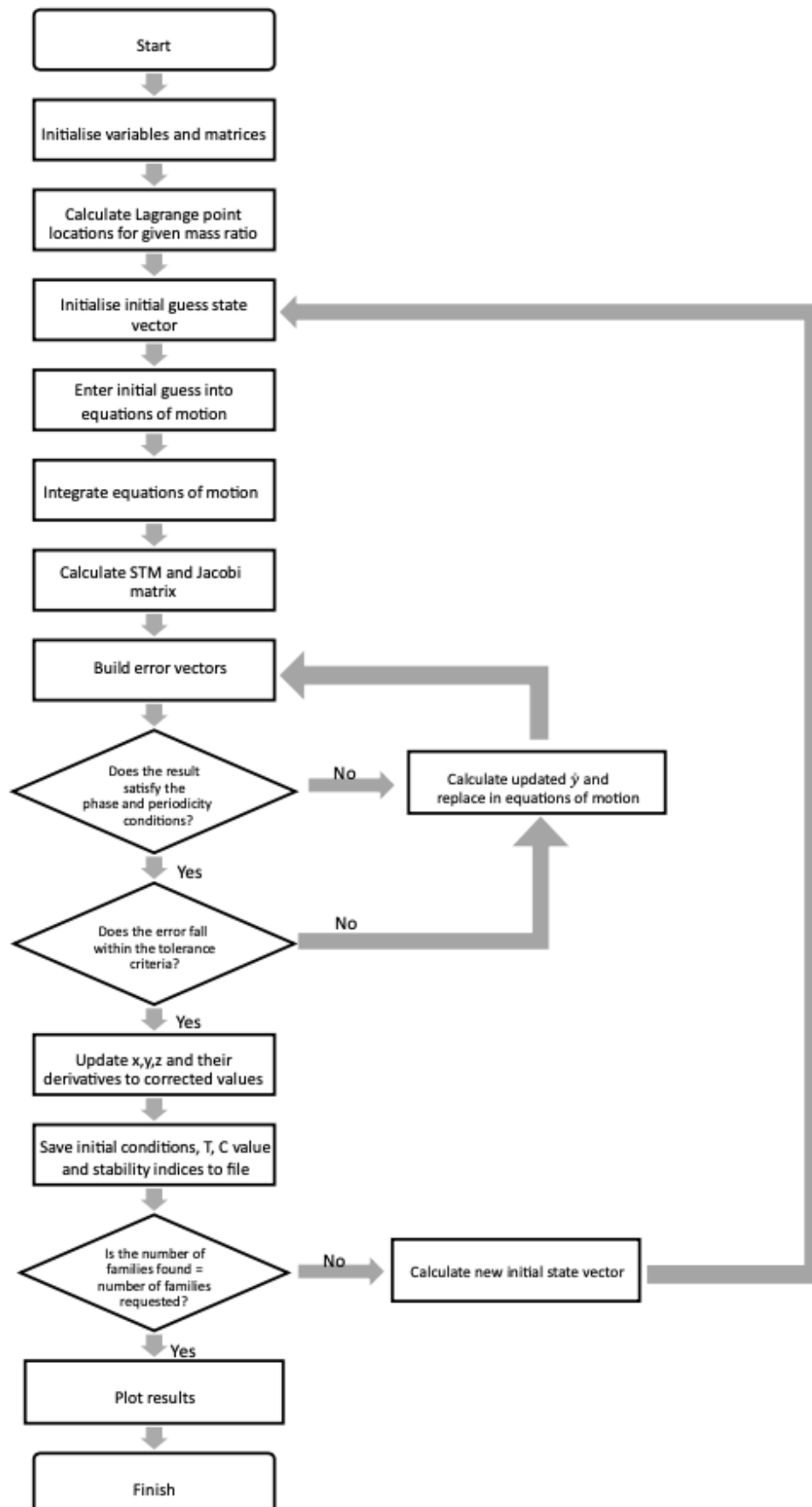


Figure 4: Flowchart representing the main program body structure, how the predictor-corrector with continuation achieves an orbital family

3.5. Lagrange Points

In accordance with the aim outlined earlier in this report, families obtained in this project will comprise periodic orbits around Lagrange points in the shortlisted dwarf planet systems. To achieve this, the Lagrange points for each system have to be located before their families can be found and so the initial stage of the algorithm employs Newton's method to locate these solutions. Newton's method finds the roots of a real-valued function, given that Lagrange points denote the equilibrium solutions to the CRTBP, this algorithm proves highly efficient and highly accurate for their identification.

Implementing this approach means an initial guess that can be refined must be provided, to do this the non-dimensional nature of the problem can be exploited. The location of the L_4, L_5 points in non-dimensional coordinates is given by

$$L_4 = \mu^* - \frac{1}{2}, \frac{\sqrt{3}}{2} \quad L_5 = \mu^* - \frac{1}{2}, \frac{-\sqrt{3}}{2} \quad (17)$$

Thus these can be input as initial guesses for the equilateral points as μ^* is known. The remaining Lagrange points are the co-linear ones, resulting in there only being one non-zero component of the state vector $\mathbf{X} = [x \ y \ z \ \dot{x} \ \dot{y} \ \dot{z}] = [x \ 0 \ 0 \ 0 \ 0 \ 0]$. L_1 lies between the primary and secondary, if μ^* is the secondary position then an initial guess for L_1 can be given by $n * [\mu^* \ 0 \ 0 \ 0 \ 0 \ 0]$, where n is a positive integer with magnitude less than 1. For L_2 the magnitude of n should be greater than 1 as this will place the initial guess to the right of the secondary mass, which is the region in which L_2 is known to be located. L_3 is to the left of the primary mass meaning an initial guess for its location can also be given by $n * [\mu^* \ 0 \ 0 \ 0 \ 0 \ 0]$ but in this case n is a negative integer. An alternative way for computing the co-linear Lagrange points is to set up a quintic polynomial through reduction of the equations of motion but this would be a lot slower and potentially less accurate depending on how the solutions are found.

3.6.Obtaining a Family

3.6.1 Implementing the Predictor-Corrector

With Lagrange point location determined, the first member in a family can be found. Planar Lyapunov orbits, as the name suggests, restrict motion of the third body to the x-y plane which incidentally reduces problem complexity in terms of initial vector and boundary constraints making them the best candidate to start with. This problem simplification was also particularly helpful at this stage as this was the initial implementation of the predictor-corrector. Given that the algorithm had been written but not rigorously tested, the likelihood of encountering debugging and troubleshooting tasks was high, reducing problem dimension minimises this potential for error. In addition to this, for CRTBP there exists symmetry in the x-y plane which means if a

$y = 0$ plane

$$x(-t), y(-t), z(-t)$$

$$x(t), y(t), z(t)$$

Figure 5: Symmetry property of the $y = 0$ plane

trajectory flows forward from the $y = 0$ plane, there will be a mirror image of this trajectory traced in the backwards direction. These end up being two parts of the same trajectory and this is a fact that can be exploited to reduce computation time when computing periodic orbits. If a trajectory is assumed to cross the $y = 0$ plane twice in a perpendicular direction then the trajectory will mirror itself before that final crossing which returns it to the state at first crossing, this will eventually create a closed (plane-symmetric periodic orbit) trajectory. A sentiment conveyed in Figure 5.

Periodic orbits by definition are trajectories that repeat themselves over a set time period, if $\mathbf{X}_0$ is the initial state vector ($\mathbf{X}_0 = [x_0 \ y_0 \ z_0 \ \dot{x}_0 \ \dot{y}_0 \ \dot{z}_0]$) and T is the time of flight of a spacecraft then a periodic orbit is found by identifying a set of variables that fulfil periodicity and phase conditions. This set of variables can be denoted as $X \equiv [\mathbf{X}_0, T]^T \in \mathbb{R}^{7 \times 1}$. As its name suggests, a periodicity condition is set to obtain an orbit which has an initial state that coincidences with the final state when propagated over time T , mathematically this can be expressed as

$$\varphi(\mathbf{X}_0, 0, T) - \mathbf{X}_0 = 0 \in \mathbb{R}^{6 \times 1} \quad (18)$$

Where $\varphi(\mathbf{X}_0, 0, T)$ is the final state vector of $\mathbf{X}_0$ propagated from 0 to T . The phase condition is implemented to prevent $\mathbf{X}_0$ from being found more than once on the periodic orbit. For the predictor-corrector, these two constraints can be written as

$$F(\mathbf{X}) = \begin{bmatrix} \varphi(\mathbf{X}_0, 0, T) - \mathbf{X}_0 \\ P(\mathbf{X}_0) \end{bmatrix} = 0 \in \mathbb{R}^{7 \times 1} \quad (19)$$

With $F(\mathbf{X})$ being the solution.

Explained in the literature review, PALC applies constraints to the guess for the solution based upon the tangential vector to the current solution and step size,

$$(\mathbf{X}_g - \tilde{\mathbf{X}})^T \mathbf{X}' - \delta s = 0 \quad (20)$$

Where $\mathbf{X}_g$ is the solution guess, $\tilde{\mathbf{X}}$ is the current solution, $\mathbf{X}'$ is the unit tangential vector and δs is step size. This additional constraint is therefore combined with (19) so that (21) is written into the corrector scheme

$$F(\mathbf{X}) = \begin{bmatrix} \varphi(\mathbf{X}_0, 0, T) - \mathbf{X}_0 \\ P(\mathbf{X}_0) \\ (\mathbf{X}_g - \tilde{\mathbf{X}})^T \mathbf{X}' - \delta s \end{bmatrix} = 0 \in \mathbb{R}^{7 \times 1} \quad (21)$$

The periodic solution $F(\mathbf{X})$ is found by iteratively updating a correction vector $\Delta \mathbf{X}$ until convergence criteria is met, this correction vector can be found using the reduction of the Taylor series expansion about the next guess $\mathbf{X}_g$ made in PALC. For the next guess, the Taylor series expansion is

$$F(\mathbf{X}_g + \Delta \mathbf{X}) = F(\mathbf{X}_g) + \frac{\partial F(\mathbf{X}_g)}{\partial \mathbf{X}_g} \Delta \mathbf{X} + H.O.T \quad (22)$$

Which, when neglecting higher order terms produces the correction vector equation

$$\Delta \mathbf{X} = - \left(\frac{\partial F(\mathbf{X}_g)}{\partial \mathbf{X}_g} \right)^{-1} F(\mathbf{X}_g) \quad (23)$$

Where $\frac{\partial F(\mathbf{X}_g)}{\partial \mathbf{X}_g}$ is the Jacobian of the system,

$$\frac{\partial F(\mathbf{X}_g)}{\partial \mathbf{X}_g} = \begin{bmatrix} \frac{\partial \varphi}{\partial \mathbf{x}_0} & \dot{\varphi}(\mathbf{x}_0, 0, T) \\ \mathbf{X}'^T_{\mathbf{x}_0} & \mathbf{X}'^T_T \end{bmatrix} \quad (24)$$

and $\frac{\partial \varphi}{\partial x_0}$ is the STM.

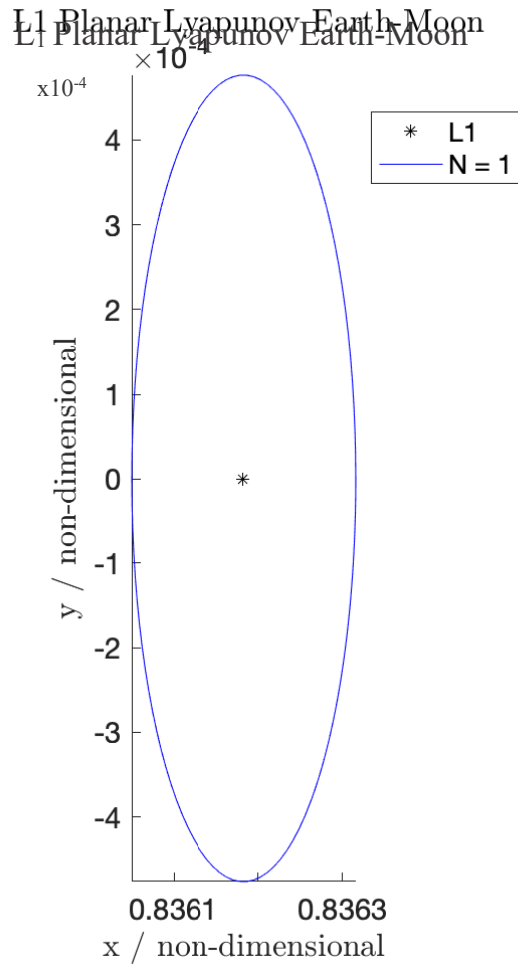


Figure 6: The first member of the Earth-Moon L_1 Planar Lyapunov family in non-dimensional units, where $\mu^* = 0.0121505856$

Applying PALC followed by single shooting to the CRTBP using this initial guess, an initial periodic orbit can be obtained. This is shown in Figure 6 which is the initial orbit for the L_1 Planar Lyapunov family in the Earth-Moon system, the system has been kept non-dimensional to demonstrate the step magnitude for continuation as discussed in the following paragraph.

To achieve the desired result of obtaining manifold families of solutions not just a singular orbit, once an initial orbit is found a continuation process is employed, making small adjustments to the initial conditions, and then re-initialising the predictor-corrector scheme to find the next member. This process is repeated systematically to trace out families of periodic orbits. For the first member of the family, like in Figure 6, the starting point is the Lagrange point and the increment will be taken relative to this. To identify what this step increment should be, the six eigenvalues and corresponding eigenvectors of the Jacobi matrix for the current Lagrange point are found. Then out of

these six, the combination which corresponds to the stable manifold with an eigenvector taking the form $[x \ 0 \ 0 \ 0 \ \dot{y}_0 \ 0]$ is selected, this eigenvector is multiplied by a step length parameter then added to the Lagrange point state vector which only has x as a non-zero term, creating the initial condition of a position offset from L_j along the x -axis with only a perpendicular velocity.

$$\mathbf{X}_0 = \tilde{\mathbf{X}}_0 + \delta s \left(\frac{\partial \tilde{\mathbf{X}}_0}{\partial s} \right) \quad (25)$$

(25) is the equation used to construct this initial guess when implemented in the algorithm, $\mathbf{X}_0$ is the initial guess, $\tilde{\mathbf{X}}_0$ is the Lagrange state vector (or position of the initial condition of the solved orbit, used as the starting point for building the initial guess of the next orbit family member in the continuation method), δs is the step size and $\frac{\partial \tilde{\mathbf{X}}_0}{\partial s}$ is the direction vector for the step determined by the eigenvalue analysis.

3.6.2 Adaptive Step Sizing

In numerical methods, step size refers to how the size of intervals over which calculations are performed is chosen during the integration process and can be either constant or adaptive. In the context of computational methods used in this project, step size refers to the distance between initial states of family members for the continuation method outlined above. As the name suggests, adaptive step size adapts based on solution behaviour, monitoring error in the solution and adjusting step size accordingly. If the error is small then larger steps are taken to speed up computation, conversely, if error is large then smaller steps are taken to improve accuracy. Despite being more complex to implement, due to its dynamic nature adaptive step sizing maintains accuracy whilst reducing computational effort which makes it a more efficient choice. Additionally, this adaptive behaviour makes this approach suitable for problems where solution behaviour is sensitive to parameter change which means it handles the sensitivity of CRTBP to parameters such as mass ratio very well. This robustness can also be exploited in the higher fidelity model as perturbations will alter the parameter space and solution behaviour within this.

To determine the appropriate step size, the approach outlined by [27] will be implemented in this project, this approach uses a criteria based on the optimal number of iterations for the current solution behaviour. Firstly, an initial step length alongside a maximum step size (Δs_{MAX}) is decided, also introduced is the optimal number of iterations for the corrector being used. If N_j is the current number of iterations and N_{opt} is the optimal number then a factor of their ratio can be used for evaluation

$$\xi = N_{opt}/N_j \quad (26)$$

Mentioned in the previous sentence, application of this factor will vary depending on the numerical method being employed, the limits chosen here are because of the predictor-corrector scheme applied for this project. If, for a given iteration $\xi > \Delta s_{MAX}$ then the step size Δs , is capped at the maximum step $\Delta s = \Delta s_{MAX} = 2$. Conversely if $\xi < \Delta s_{MIN}$ then $\Delta s = \Delta s_{MIN} = 0.5$, finally if ξ is found to be within these boundaries then $\Delta s = \text{minimum}(\xi \Delta s, \Delta s_{MAX})$ which means the step size is taken to be the minimum of these two options.

With adaptive step size incorporated into the continuation method, families of orbits could be obtained, Figure 7 below depicts the application of continuation to the initial orbit obtained in Figure 6. Notably, the alternating regions of orbit bunching and spacing directly arise from the utilisation of the adaptive approach.

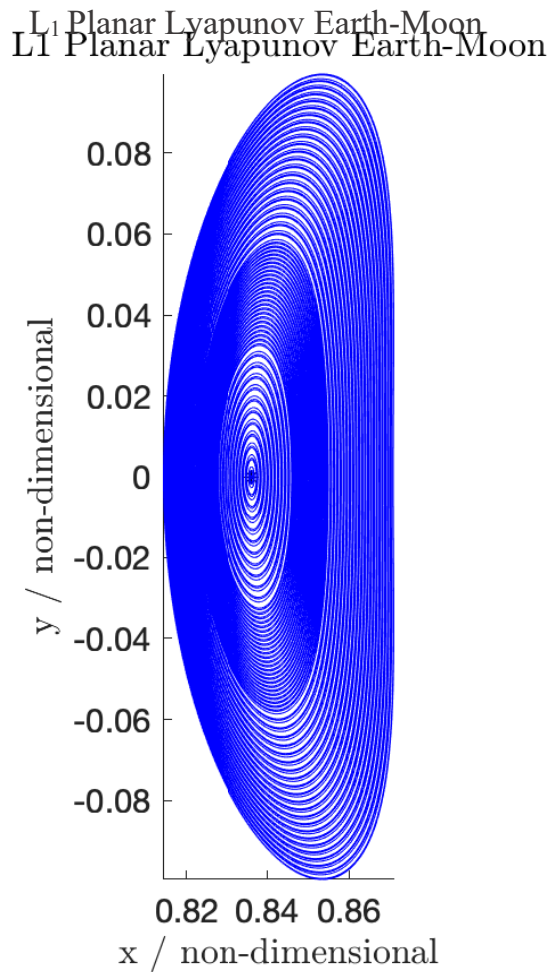


Figure 7: L_1 Planar Lyapunov family for the Earth-Moon system where $\mu^* = 0.0121505856$. Results shown are for 400 iterations and are presented in non-dimensional units.

3.6.3 Validation

An example of the validation procedure put in place is shown in the comparison between the families in Figure 8. The family obtained by the predictor-corrector has a C value range from 3.1767-3.1897 and if this range is inputted into the NASA tool it produces the right-hand plot in Figure 8. A quick note is NASA's tool takes the planet, in this case Earth, to be in the negative x direction hence why the axes have strange numeration. The focus here is on the shapes of the families, both taking on the kidney bean shape therefore the predictor-corrector and continuation methods are working as desired.

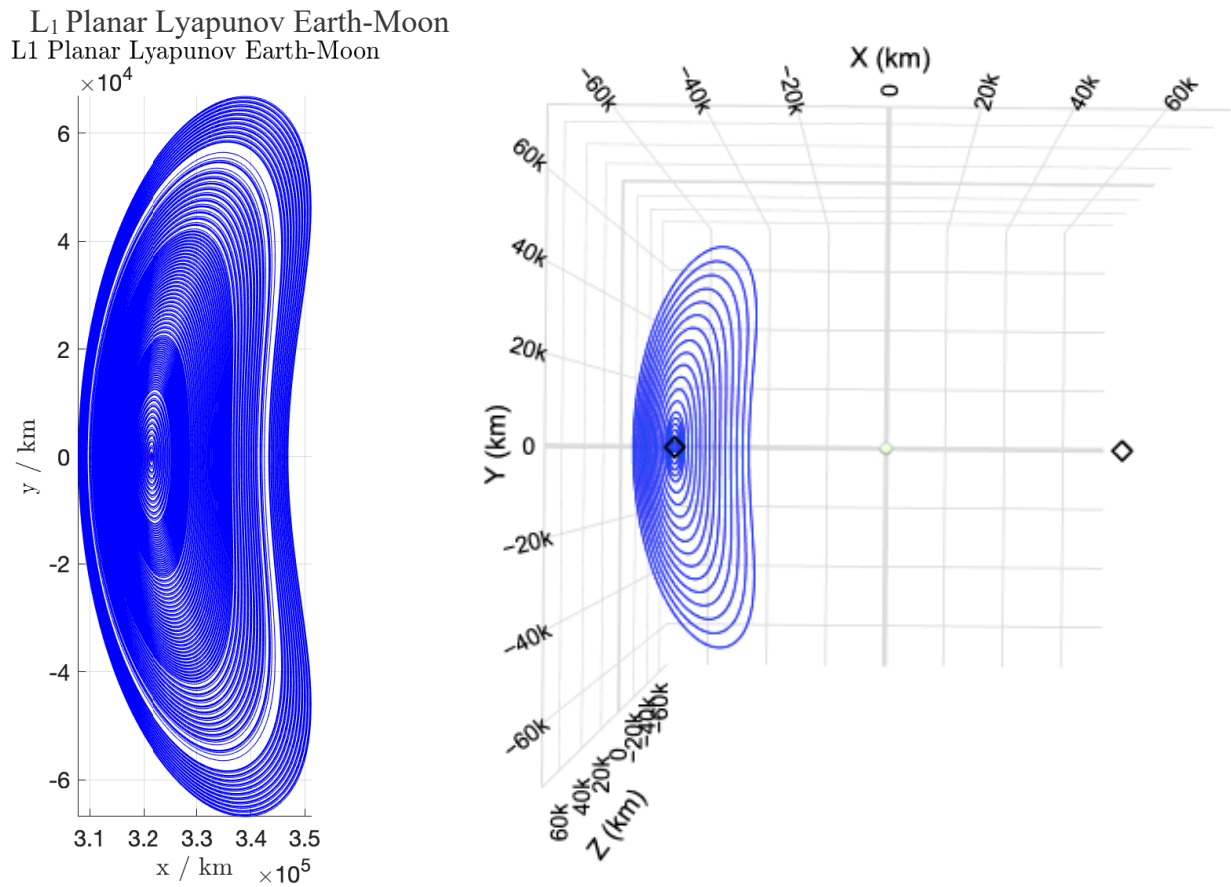


Figure 8: Comparison between the results obtained by the predictor-corrector (left) for the Earth-Moon L_1 Planar Lyapunov family and the precomputed orbits available in the NASA database tool (right [12]). The predictor-corrector family spans a smaller distance (31k-35k) but this is due to the number of iterations, further continuation can be applied to solve this.

3.6.4 Application to Dwarf Planet Systems

Proceeding on to satisfying the aim of this project, identifying periodic orbits around Lagrange points for dwarf planet systems, Figure 9 is the result of changing the mass parameter to be that of the Haumea-Hi'iaka system where $\mu^* = 0.00447984071$. Visible in the similarities between Figure 7 and Figure 9, the families of orbits for each system will look almost identical and so including figures of every family for every Lagrange point

in every system in the main body of this report would be redundant. That being said, these are the results of this project and so for each family found, one case will be displayed in the main body and the remaining examples can be found in appendix I of this report. On a similar note, it should be noted that the Planar Lyapunov family only exists for co-linear Lagrange points and so there will be no figures for the equilateral points in this case.

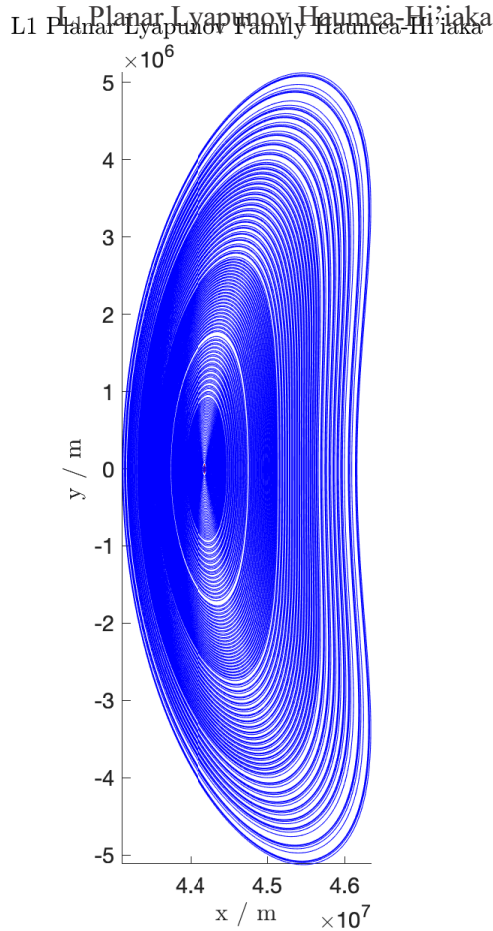


Figure 9: Results of applying the predictor-corrector and continuation methods around the L_1 point when

3.7. Bifurcation Theory and the Halo Family

Halo orbits can be found by identifying tangent bifurcation points in Planar Lyapunov families and then propagating out of plane in the z -direction, with positive z providing the Northern family and negative z providing the Southern family. As outlined in section 2.6 of the literature review, this critical bifurcation point can be found by identifying when the stability index, $b_j = 2$ as this corresponds to the transition between a stable and unstable orbit.

3.7.1 Bisection Method

For every member in an orbital family, the predictor-corrector method implemented in this computer model not only calculates and outputs the member's initial conditions but also finds the corresponding C value, time period and stability indices; the latter being calculated by equation 16 from section 2.6. A search through the list of indices is insufficient for identification of the critical point as it is extremely unlikely there will be an exact value of 2 in this list, instead a bisection method can be used.

This bisection function was created in a separate script to maintain readability of the code and to ensure the integrity of the predictor-corrector remains unaffected by any unintended interaction from this function. Hence, it begins with loading in the data for the current family being analysed; this data comprising of the state vectors, time period, C value and stability indices for each orbital member in the family. In this model, b_1 was the index used in the bisection method and so will be the stability index referred to hereafter, but b_2 would also be applicable. A variable is created for the 'desired index' b_{des} , in this case $b_{des} = 2$, and then the algorithm performs a search of b_1 values loaded in from the file sorting them into two lists, those greater than b_{des} and those less than b_{des} . After this, the member closest to b_{des} is selected from each list and this becomes the range for the initial bisection. This function is an iterative process that works by taking the midpoint of the range and then evaluating a parameter at this point, based on this value the range is updated. In this case, the parameter will be b_1 , if $b_{mid} < b_{des}$ the lower bound of the range is updated to the midpoint, conversely when $b_{mid} > b_{des}$ the upper bound of the range is updated; this process essentially narrows down the range where the desired index is located.

As with all numerical methods, iteration continues until the error falls within a given tolerance and when this occurs, the current value becomes the solution. Given that the bisection method isn't computationally intense and accuracy is vital due to the innate instability associated with complex dynamics, the tolerance was set to 10^{-20} , so the method will stop when $b_{mid} = 2 \pm 10^{-20}$. Simultaneous to the bisection of stability index, the other parameters are undergoing bisection too, they are just irrelevant to the location of critical point so are excluded from the comparison stage. The result of this is that when the process terminates there is a state vector, time period and C value corresponding to the bifurcation point and this is a set of initial conditions to be used for obtaining the Halo family bifurcating from a given Planar family. For further clarification, a flowchart representing this bisection method can be found in appendix II.

Figure 10 is the stability diagram obtained for L_1 Planar Lyapunov family in the Earth-Moon system, using the data from the family in Figure 7 as the bisection method input. The shaded region is indicative of where the orbit transitions to unstable.

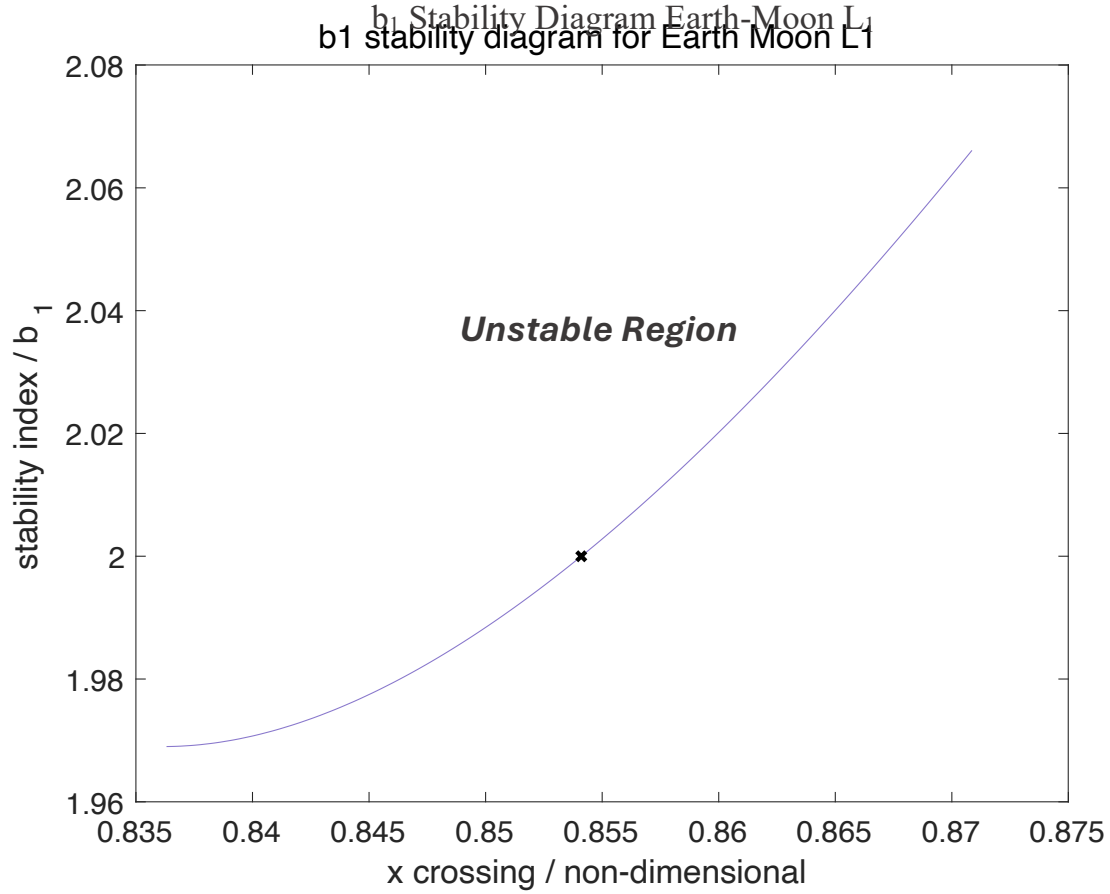


Figure 10: Stability diagram for the Earth-Moon L_1 Planar Lyapunov family found in section 3.6. Shaded region is indicative of the transition to unstable orbits, as expected the bifurcation occurs at this transition

Also to emphasise here is that the critical value, b_{des} , was chosen to be 2 as this corresponds to the tangent bifurcation which can be used for obtaining Halo orbits for the stability indices used in this project. Due to the fact the CRTBP has been thoroughly explored there are a multitude of approaches in solving it, with a multitude of ways to calculate parameters to measure the results, these parameters include stability indices.

3.7.2 Halo Orbits

Using the initial condition found as a result of the bisection method, Halo orbits corresponding to L_1 in the Earth-Moon system can be found by propagating in the positive and negative directions for the Northern and Southern families respectively.

For the Planar Lyapunov family, the initial guess is constructed using an initial position and a direction vector determined by eigenvalue decomposition of the Jacobian at the

Lagrange points. As discussed, the bisection method identifies the initial position for the Halo family guess however the step direction still remains to be determined. Instead of applying eigenvalue decomposition, a method called Singular Value Decomposition (SVD) was used. For a given matrix A , SVD will decompose this into three smaller matrices, a unitary matrix containing all the left singular vectors, another with the right singular vectors and finally a diagonal matrix containing the singular values, the singular vectors can be used for identifying the stable direction for propagation. Both approaches achieve the same outcome, an insight into dynamics and stability, SVD becomes more applicable here as the decomposition can provide a clearer understanding of underlying dynamics in systems with sensitive bifurcation structures and multiple unstable eigenvalues. As with Planar Lyapunov orbits the components are analysed and in this case, the one which corresponds to having a vector with its only non-zero component being $\dot{z}$ is identified. Using this for calculating step direction in the predictor-corrector, an out of plane propagation, or Halo orbit, is obtained; Figure 11 shows the Northern Halo resulting from the bifurcation analysis in 3.7.1.

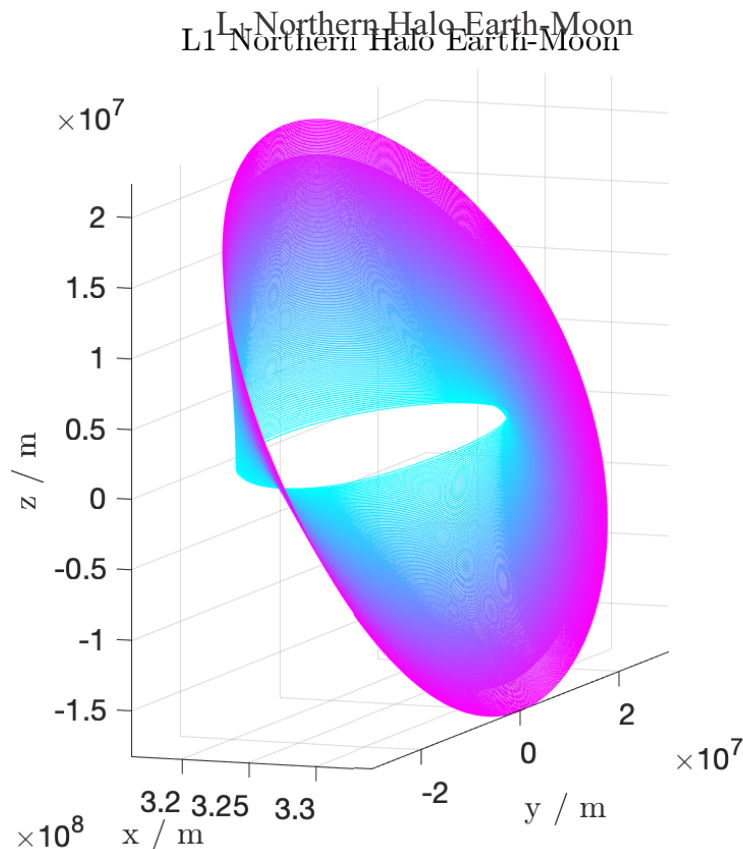


Figure 11: L_1 Northern Halo family for Earth-Moon family found in section 3.6, using the state vector found through bisection methodology as the initial condition.

Repeating the validation process, NASA's database provides precomputed orbits for the Halo family therefore a comparison of Figure 11 and NASA's results ensued. They were found to match hence the Halo family for co-linear Lagrange points in the dwarf planet systems listed in the objectives section of this report could be found. Once again, the Halo family only exists for co-linear Lagrange points so in appendix I there will only be figures for these. Figure 12 is what the Northern Halo looks like for the Eris-Dysnomia system's L_2 point where $\mu^* = 0.00488618758$.

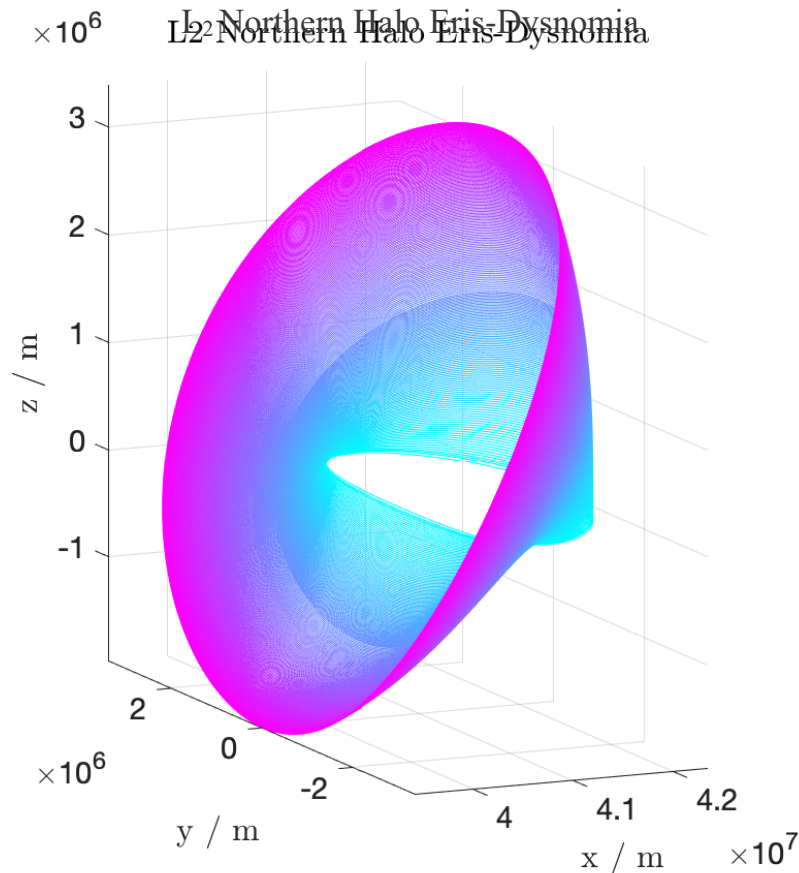


Figure 12: Northern halo family for L_2 in Eris-Dysnomia system ($\mu^* = 0.00488618758$), initial condition obtained through bisection of L_2 Planar Lyapunov family found in appendix I.

3.8. Vertical Lyapunov

Due to the nature of the dynamical equations, most orbital families in a system can be linked to one another and thus there are various ways to obtain each family; the Vertical family can be obtained through bifurcation analysis of precomputed orbits but a simpler way to obtain them is simply through identifying the correct initial eigenvector for step direction. Regressing back to the original script where Planar Lyapunov families were calculated, instead of selecting the eigenvector-value combination for a vector with a non-zero $\dot{x}$ and $\dot{y}$, the vector with only a non-zero $\dot{z}$ should be chosen for the initial

condition. If this is done, the Vertical Lyapunov family can be obtained, Figure 13 is this family for L_3 in the Earth-Moon system.

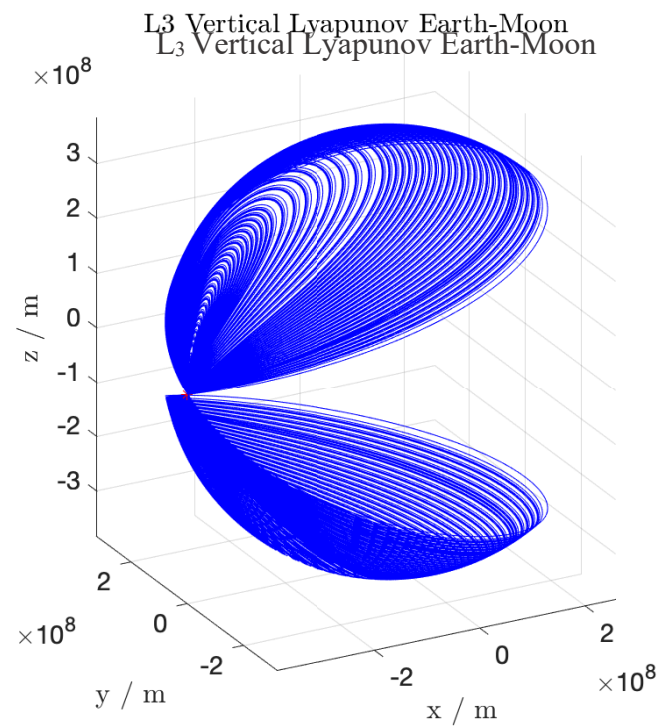


Figure 13: Vertical Lyapunov family for Earth-Moon L_3

After validating this result using the same process as discussed previously, the Vertical family was found for all systems of interest, Figure 14 shows the result for the Pluto-Charon system where $\mu^* = 0.10857221044$.

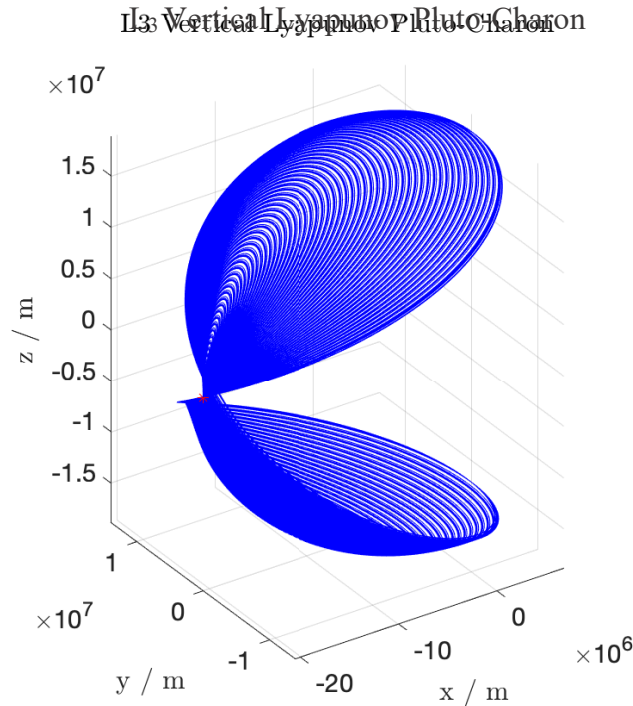


Figure 14: Vertical Lyapunov family for Pluto-Charon L_3 , $\mu^* = 0.10857221044$

3.9. Higher Fidelity Model

With the desired orbital families for the desired dwarf planet systems obtained, the low fidelity portion of this project was complete and progress into higher fidelity could begin. Mentioned in section 3.3, increasing the fidelity will be done by introducing perturbations into the CRTBP equations of motion and consequent derivatives instead of transitioning into an entirely different higher fidelity model (i.e. ERTBP). Re-iterating the previous evaluation, this was mostly done due to the time scope restraints on this project, to transition into a different problem would require an additional thorough literature review so a new computational method could be written and then this would also require validation processes before it could be applied to the desired systems. Contrastingly, introducing perturbations not only offers a more flexible approach but means the computer model used in the low fidelity approach, that has already been validated and thoroughly tested, can be used as a base – adjusting the equations of motion within it will not alter the integrity of the predictor-corrector.

Also discussed in section 3.3 was how the perturbations will vary from system to system, if the system being studied was Sun-Mercury then it would be wise to have perturbations that consider the effects of solar radiation pressure. Reaching bodies like Pluto or Haumea, even Earth, solar radiation pressure effects become almost negligible and factors like oblateness or axial tilt become predominant with regards to influence on the

orbital environment. The implementation of oblateness effects is discussed here as an illustrative example of including perturbation effects but it is important to emphasise that the changes made to the model (i.e. remodelling the derivatives) are the same for any perturbation type.

There was the option of creating perturbation equations but this would require extensive research and validation which, as explained before with other complex features, is not feasible within the time scope of this project. Therefore the equations used were sourced from reliable, published sources. In this case, the perturbed equations were sourced from [29] and have coefficients for both solar radiation pressure, q_1, q_2 , and oblateness effects A_1, A_2 where the subscript denotes if they are effects from the primary or secondary body.

The equations of motion for this perturbed problem are

$$\begin{aligned}\ddot{x} - 2n\dot{y} &= \Omega_x^* \\ \ddot{y} + 2n\dot{x} &= \Omega_y^* \\ \ddot{z} &= \Omega_z^*\end{aligned}\tag{27-29}$$

Where $\Omega^* = \frac{1}{2}n^2(x^2 + y^2) + \frac{(1-\mu)q_1}{r_1} + \frac{(1-\mu)q_1A_1}{2r_1^3} + \frac{\mu q_2}{r_2} + \frac{\mu q_2A_2}{2r_2^3}$ (30) and $n^2 = 1 + \frac{3}{2}(A_1 + A_2)$ (31)

Explained above, all of the systems considered in this study do not experience anything more than negligible solar radiation pressure due to their distance from the Sun and do not emit their own radiation therefore this radiation coefficient is redundant, to eliminate its effects $q_1 = q_2 = 1$. This reduces the potential to

$$\Omega^* = \frac{1}{2}n^2(x^2 + y^2) + \frac{(1-\mu)}{r_1} + \frac{(1-\mu)A_1}{2r_1^3} + \frac{\mu}{r_2} + \frac{\mu A_2}{2r_2^3}\tag{32}$$

With the equations of motion and potential function defined with data for the additional parameters available, this perturbation is ready to be implemented into the model. For the iteration part of the predictor-corrector, the second partial derivatives of the potential function are required which introduces a high chance of error as there is no way to validate them to be correct aside from assessing the results produced by the predictor-corrector. To minimise the chance of error, the derivatives were found both analytically and with various software, with the answers obtained across all platforms matching it was safe to assume the derivatives were correct.

The oblateness coefficient for each body can be found using

$$A = (R_e^2 - R_p^2)/5R^2 \quad (33)$$

where R_e and R_p are the equatorial and polar radii of the body respectively and R is the (average) distance between the primary and secondary.

The Earth-Moon system has very little oblateness associated with either body, the Earth's radii differ by 42km which is incredibly small relative to its entire diameter and so its oblateness coefficient is only $3.619774212e-7$, the Moon's coefficient is negligible ($\sim 10^{-9}$) thus there is very little difference seen between the perturbed and unperturbed orbits, visible in Figure 15. In order to ensure this comparison is valid, the orbits being compared were of the same C value.

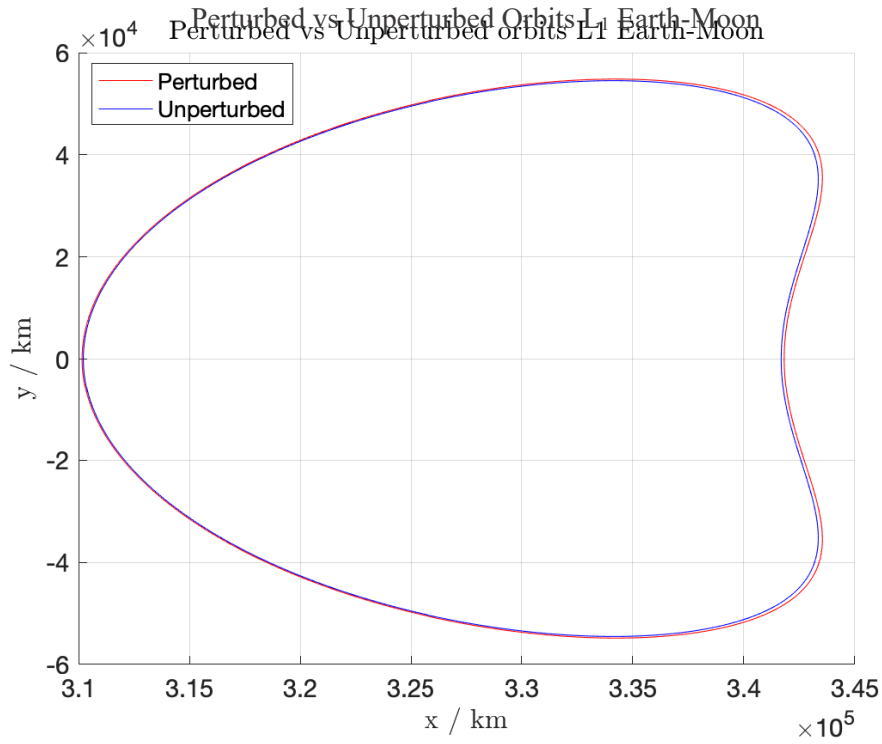


Figure 15: Comparison of the perturbed and unperturbed orbit results for the L_1 Earth-Moon Planar Lyapunov family, orbits displayed are for when $C = 3.172$.

This differs greatly to a system like Haumea-Hi'iaka. Haumea spins so quickly that it takes on the appearance of a tri-axial ellipsoid, so for Haumea

$$A = \frac{(R_e^2 - R_p^2)}{5R^2} = \frac{(1050^2 - 840^2)}{5(49880)^2} = 3.190496004e - 5$$

which is a coefficient two magnitudes larger than that of Earth. When the influence of Haumea's oblateness is considered in the orbit of a third body in the Haumea-Hi'iaka system, there is a large discrepancy between the perturbed and unperturbed orbits as seen in Figure 16.

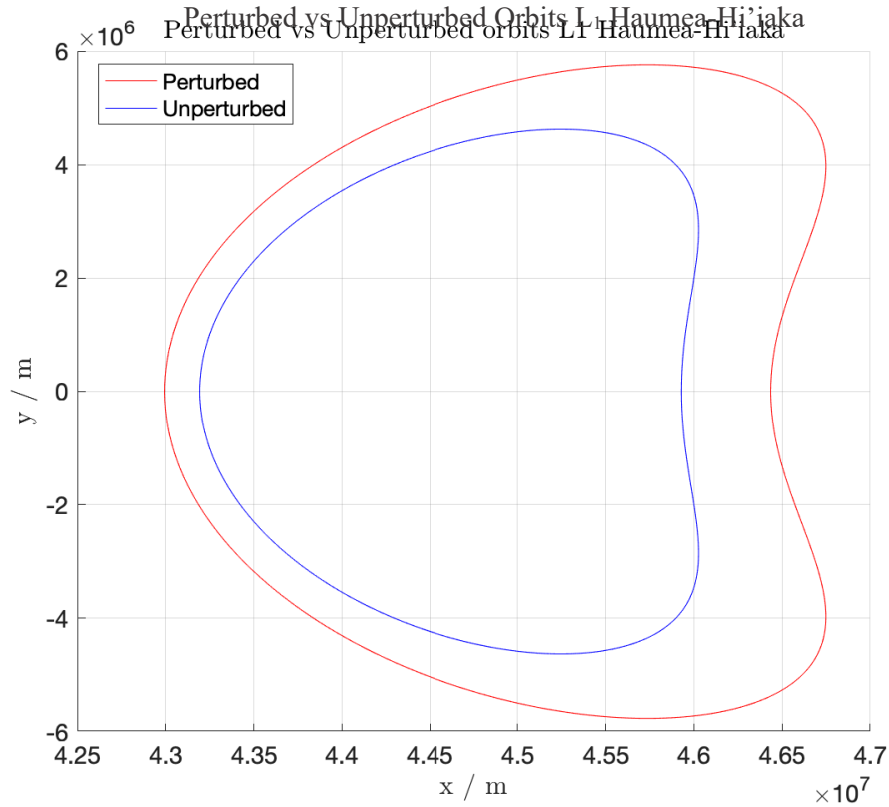


Figure 16: Comparison of the perturbed and unperturbed orbit results for the L_1 Haumea-Hi'iaka Planar Lyapunov family, orbits displayed are for when $C = 3.0946$.

The fact there are discrepancies between the results of the perturbed and unperturbed models supports the need for a higher fidelity model and that the results obtained in the CRTBP are generally correct but should only be used as a starting point for trajectory design. For conclusions about the dynamical environment in dwarf planet systems, not only can the perturbed models be used as is but can be further adapted to include multiple perturbations in each system which will further increase study accuracy and allows for the investigation to be tailored to specific mission objectives.

That being said, the initial conditions for the unperturbed dwarf planet systems remain valid, accurate results and given that not only were initial conditions but their corresponding C , time period and stability values obtained, a database like the NASA one could be built but for these dwarf planet systems. This compiled database could then be made commercially available, filling in the gap in research that currently exists and enabling others to use this work in their own research, potentially as a basic starting point for real world mission trajectories.

4. Conclusions

This project utilised a predictor-corrector algorithm alongside continuation techniques to acquire families of periodic orbits around Lagrange points within dwarf planet systems. Employing bifurcation theory facilitated the exploration of more intricate orbit families, such as the Halo families. Validation of the results was conducted through comparison with precomputed orbits based on real world data, available in NASA's three body database. Initial focus was on acquiring unperturbed results for the basic CRTBP, followed by an enhancement of model fidelity through the introduction of perturbations, resulting in observable discrepancies between the low and high-fidelity results, underscoring the necessity for more complex analyses. While unperturbed solutions should only be considered as a starting point for actual trajectory design, their utility was recognised in creating a commercially available database equivalent to NASA's but for dwarf planet systems, offering a foundational resource for future investigations into this field.

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APPENDIX I – Orbital Families

Pluto-Charon - $\mu = 0.10857221044$

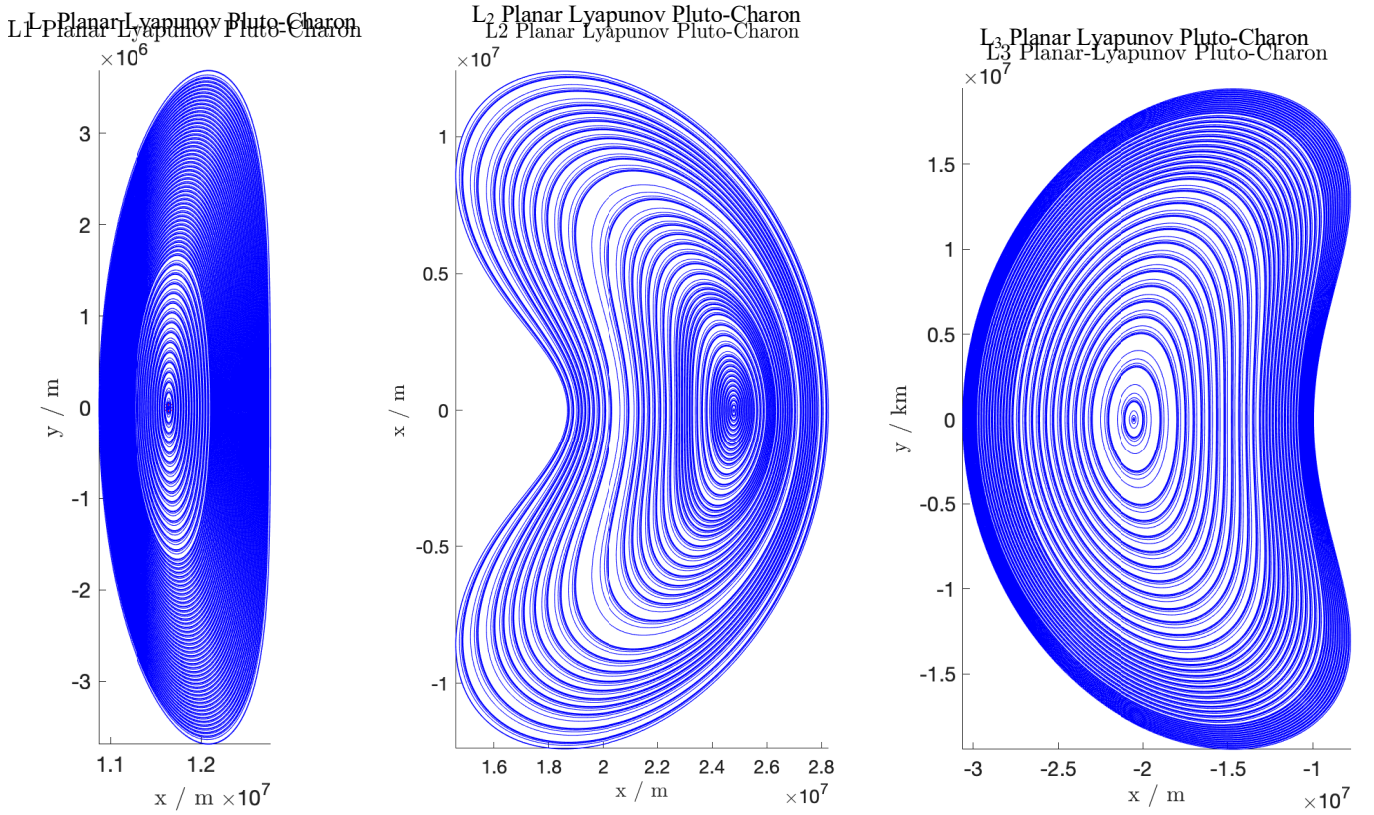


Figure 17: Planar Lyapunov family for Pluto-Charon

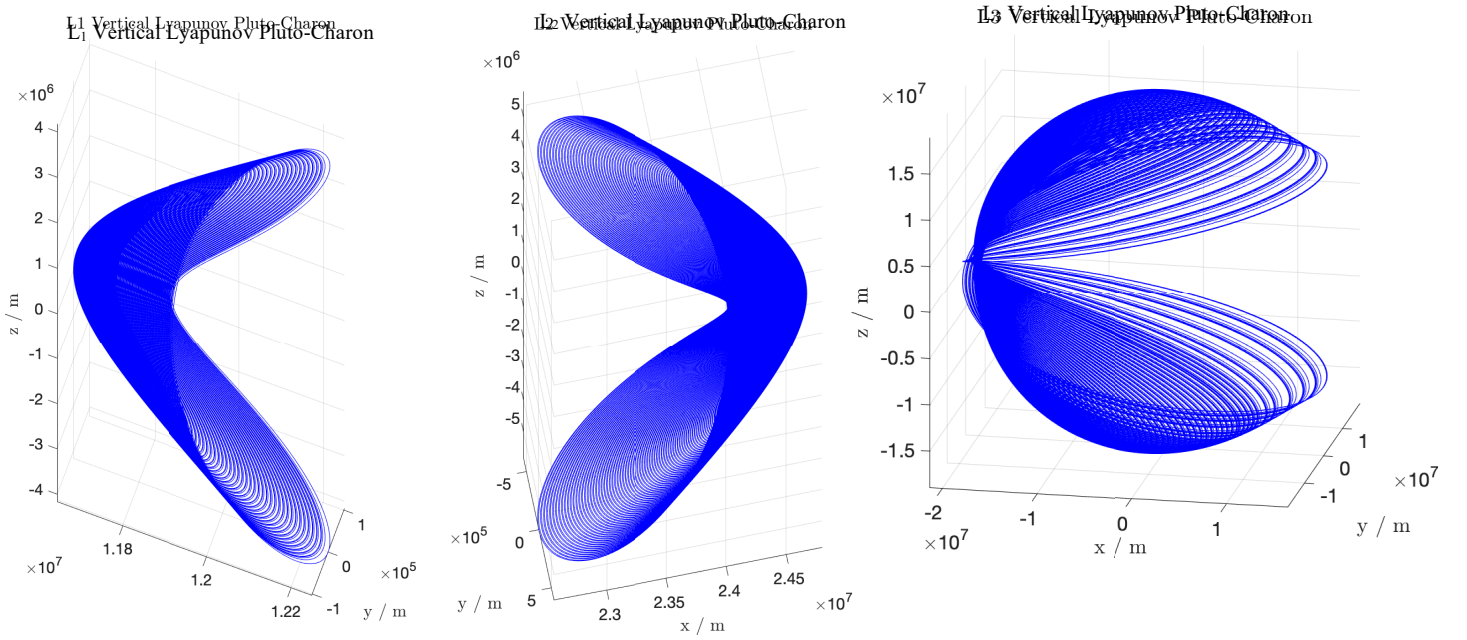


Figure 18: Vertical Lyapunov family for Pluto-Charon

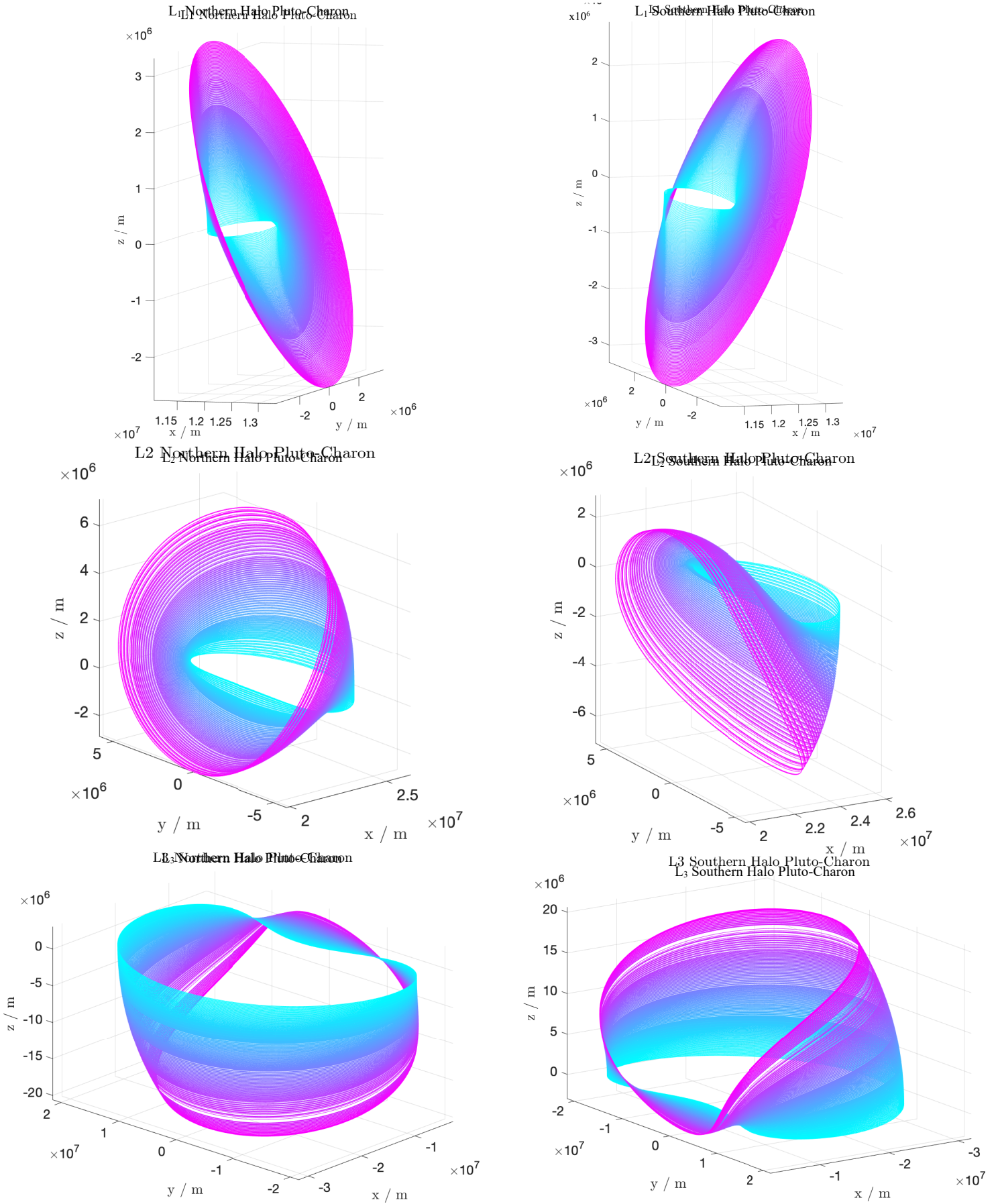


Figure 19: Northern and Southern Halo Families for Pluto-Charon

Pluto-Nix - $\mu = 0.00000345355$

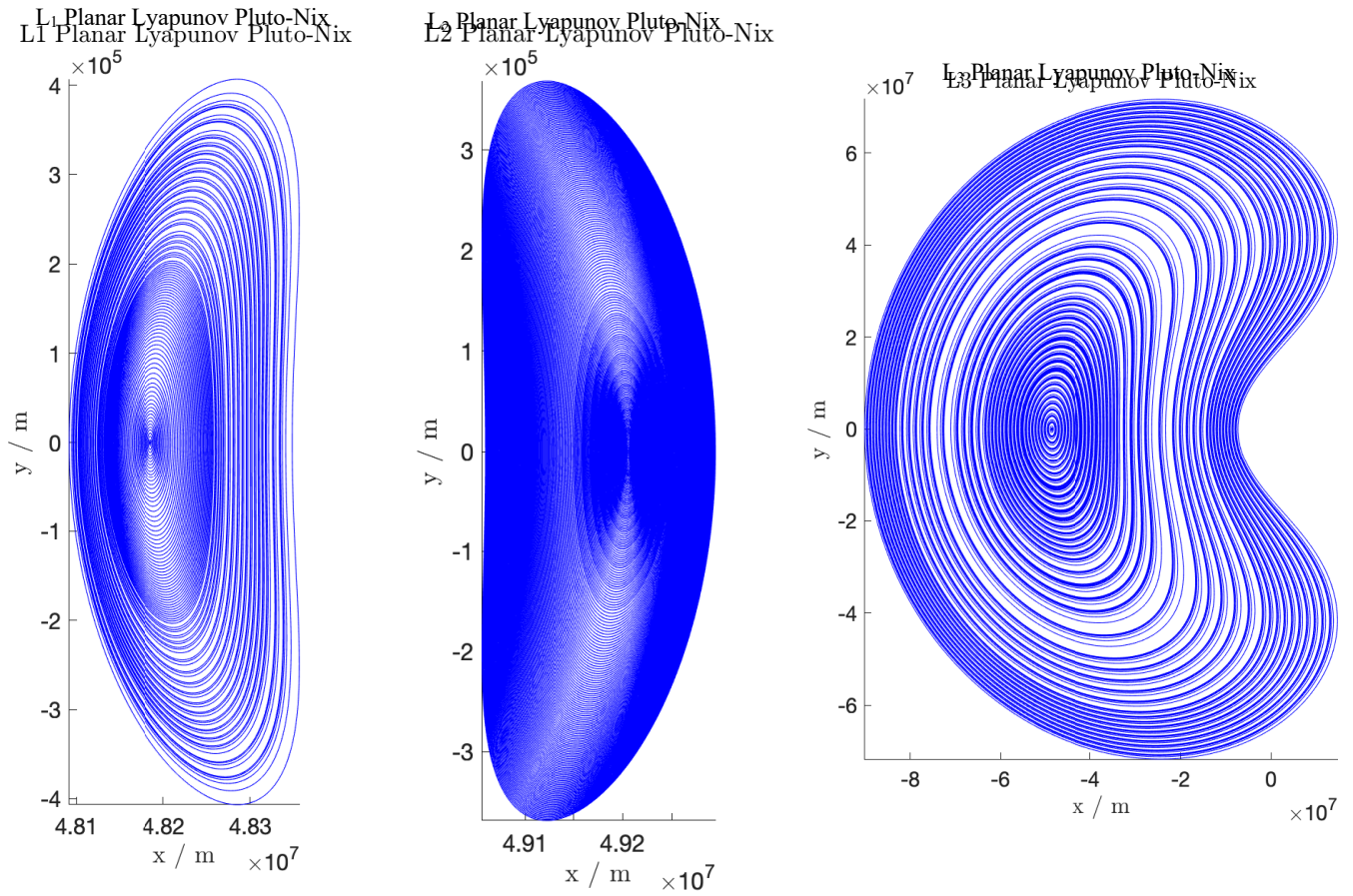


Figure 20: Planar Lyapunov family for Pluto-Nix

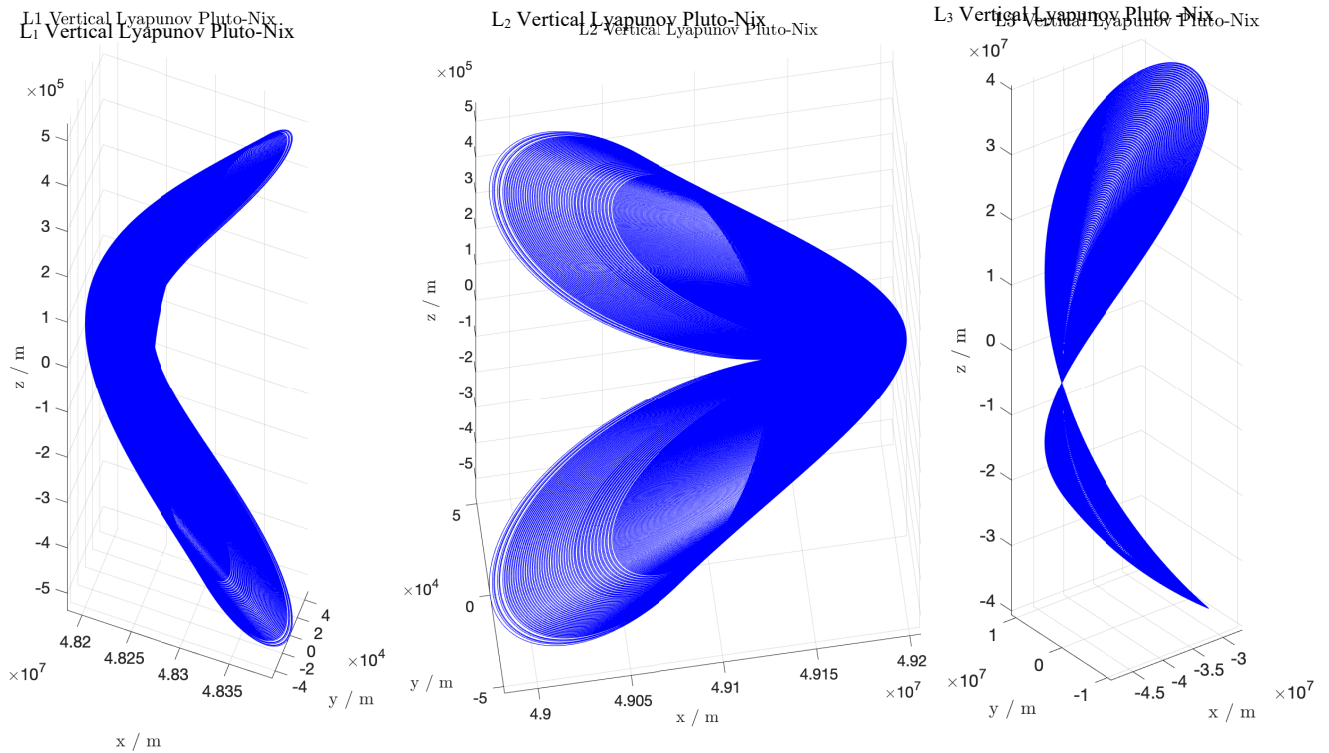


Figure 21: Vertical Lyapunov family for Pluto-Nix

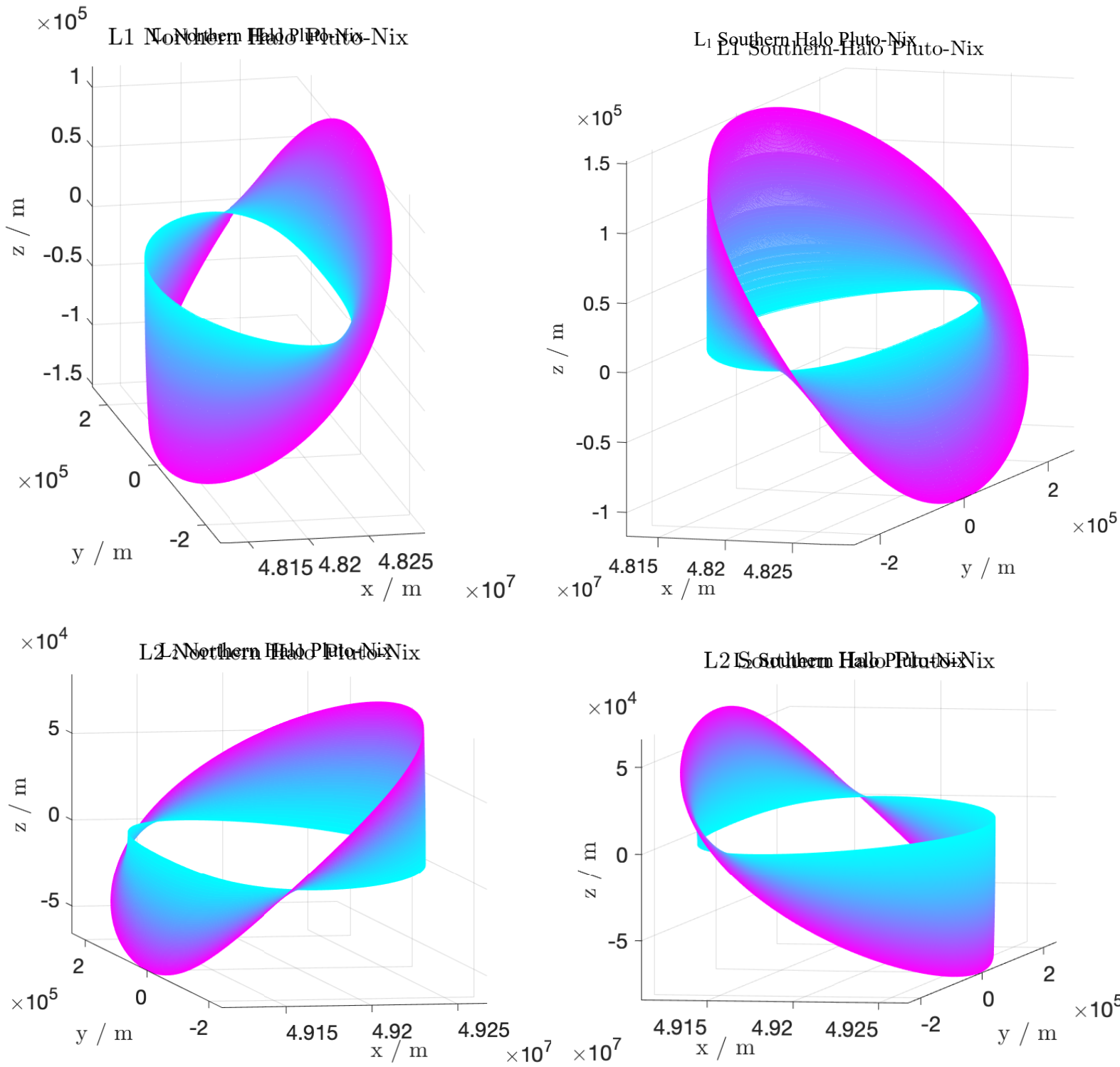


Figure 22: Northern and Southern Halo Families for Pluto-Nix

Pluto-Styx - $\mu = 0.000000575594$

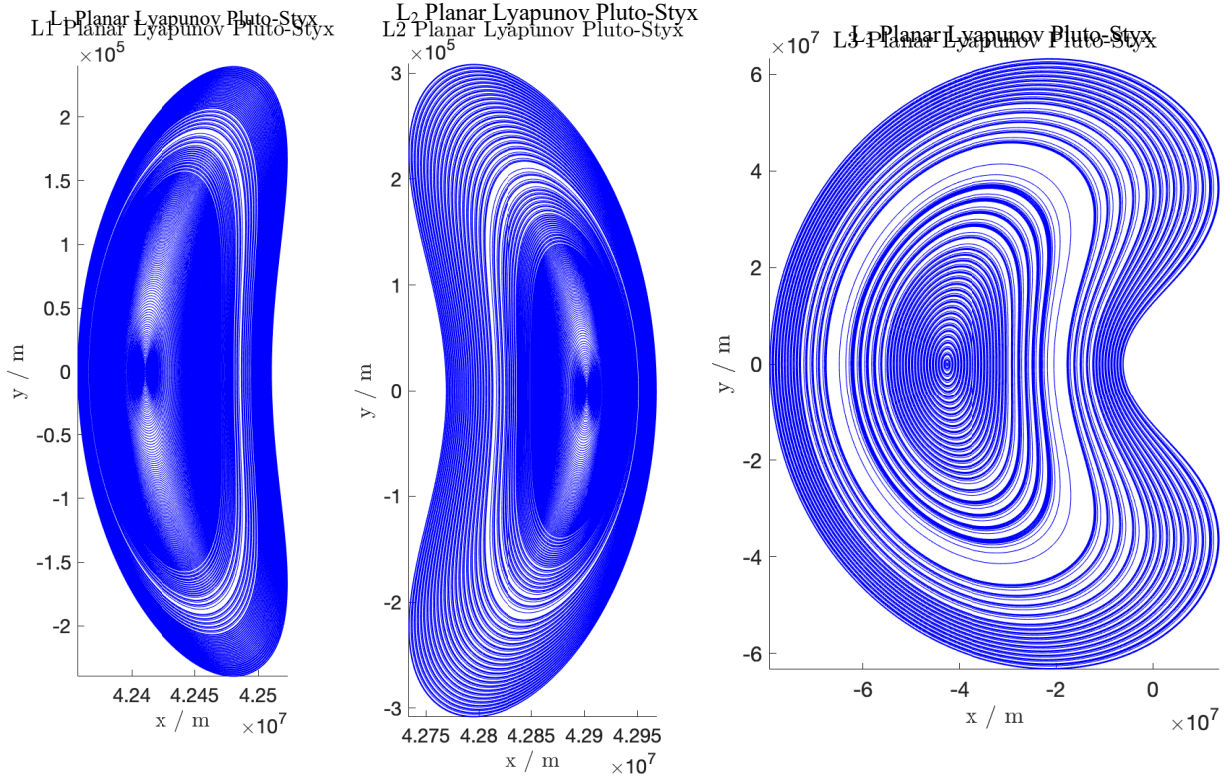


Figure 23: Planar Lyapunov family for Pluto-Styx

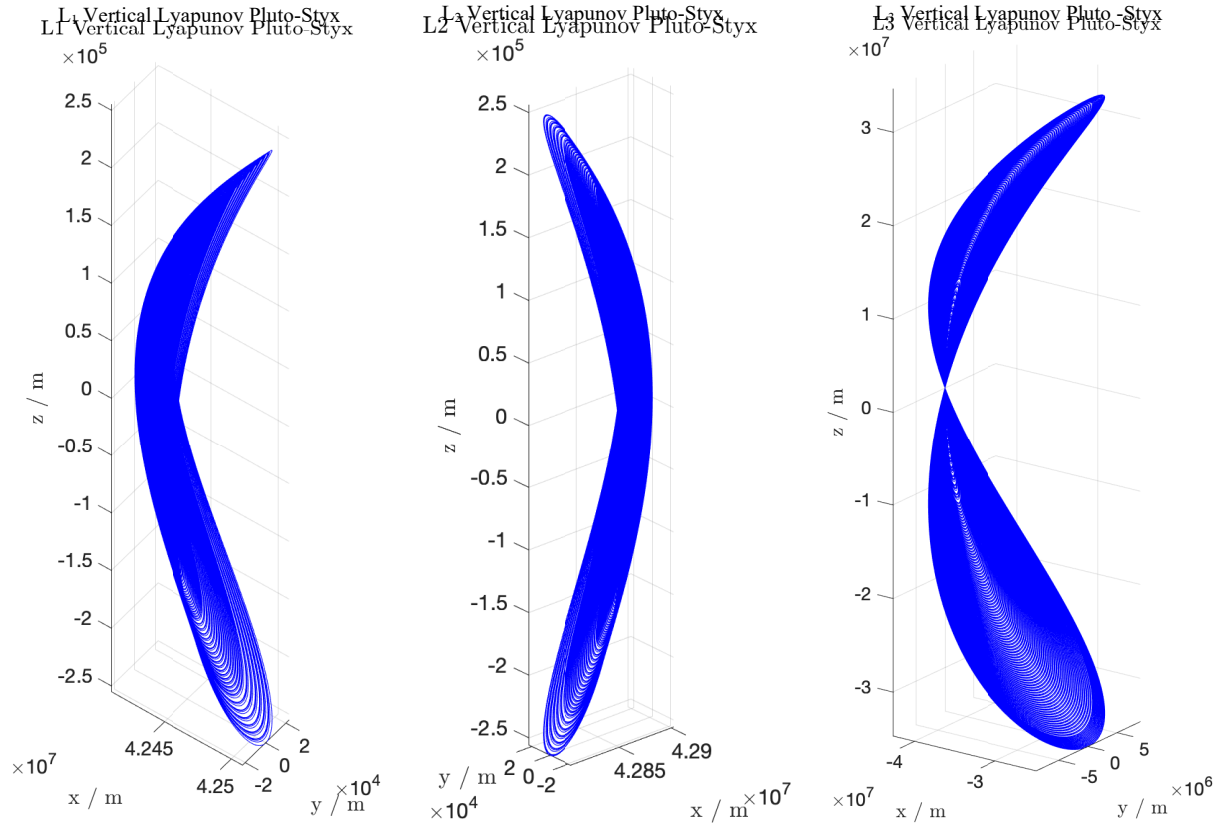


Figure 24: Vertical Lyapunov family for Pluto-Styx

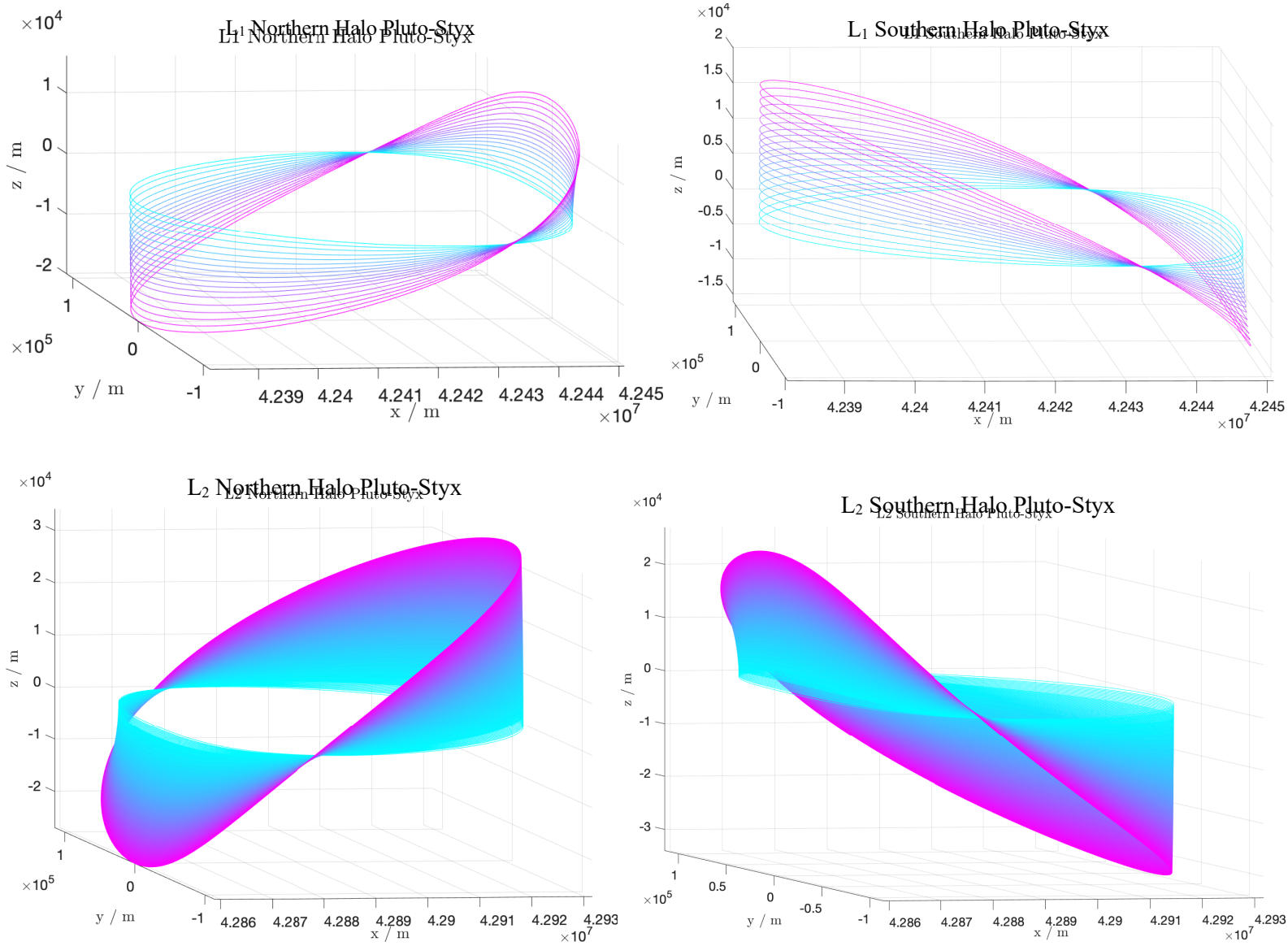


Figure 25: Northern and Southern Halo Families for Pluto-Styx

Pluto-Hydra – $\mu = 0.00000368379$

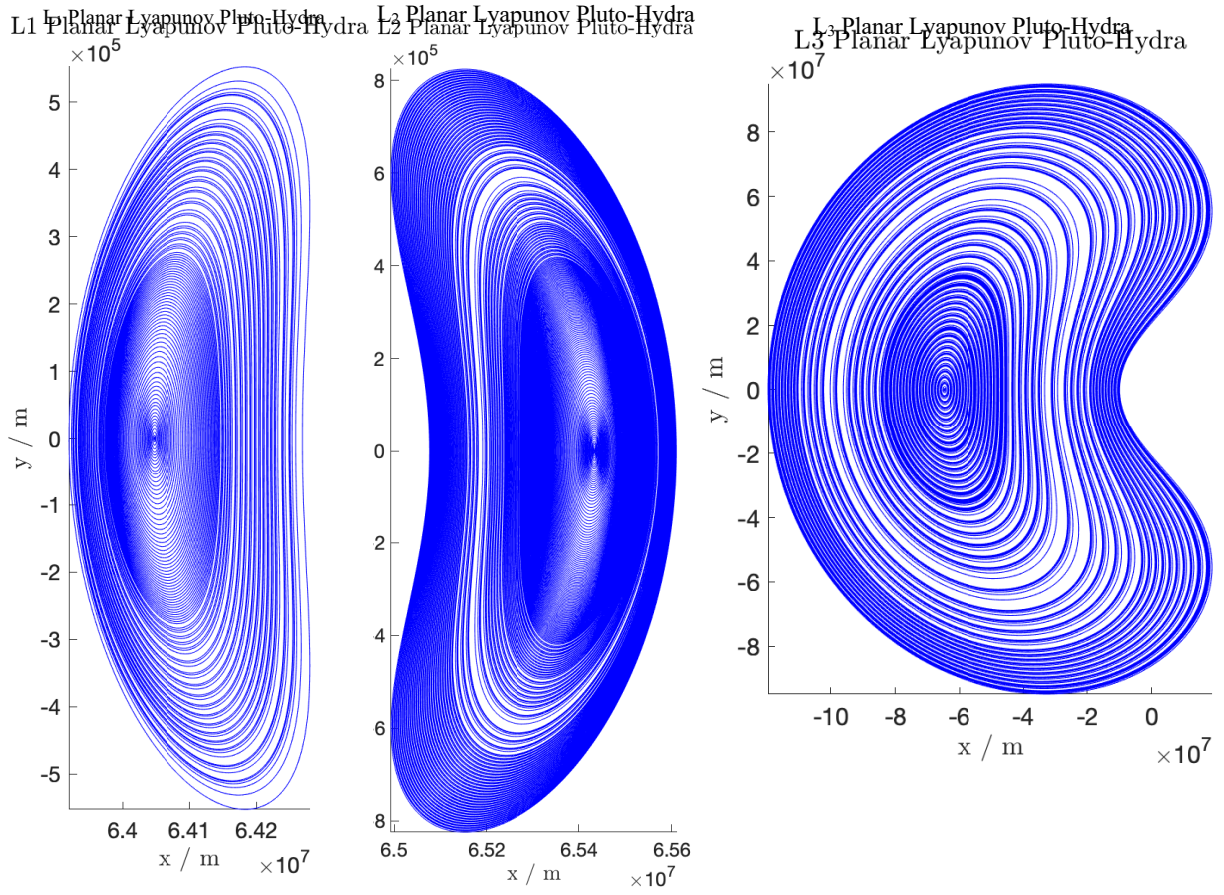


Figure 26: Planar Lyapunov family for Pluto-Hydra

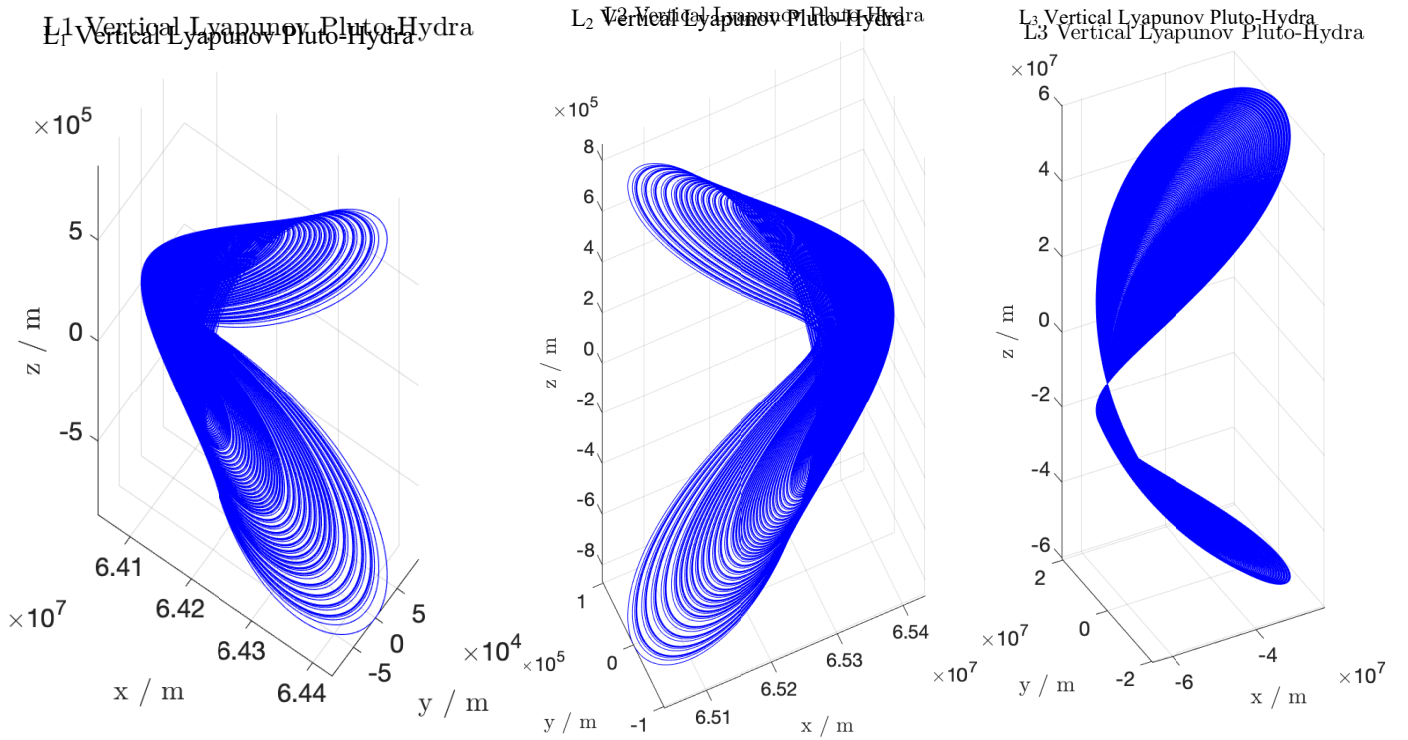


Figure 27: Vertical Lyapunov family for Pluto-Hydra

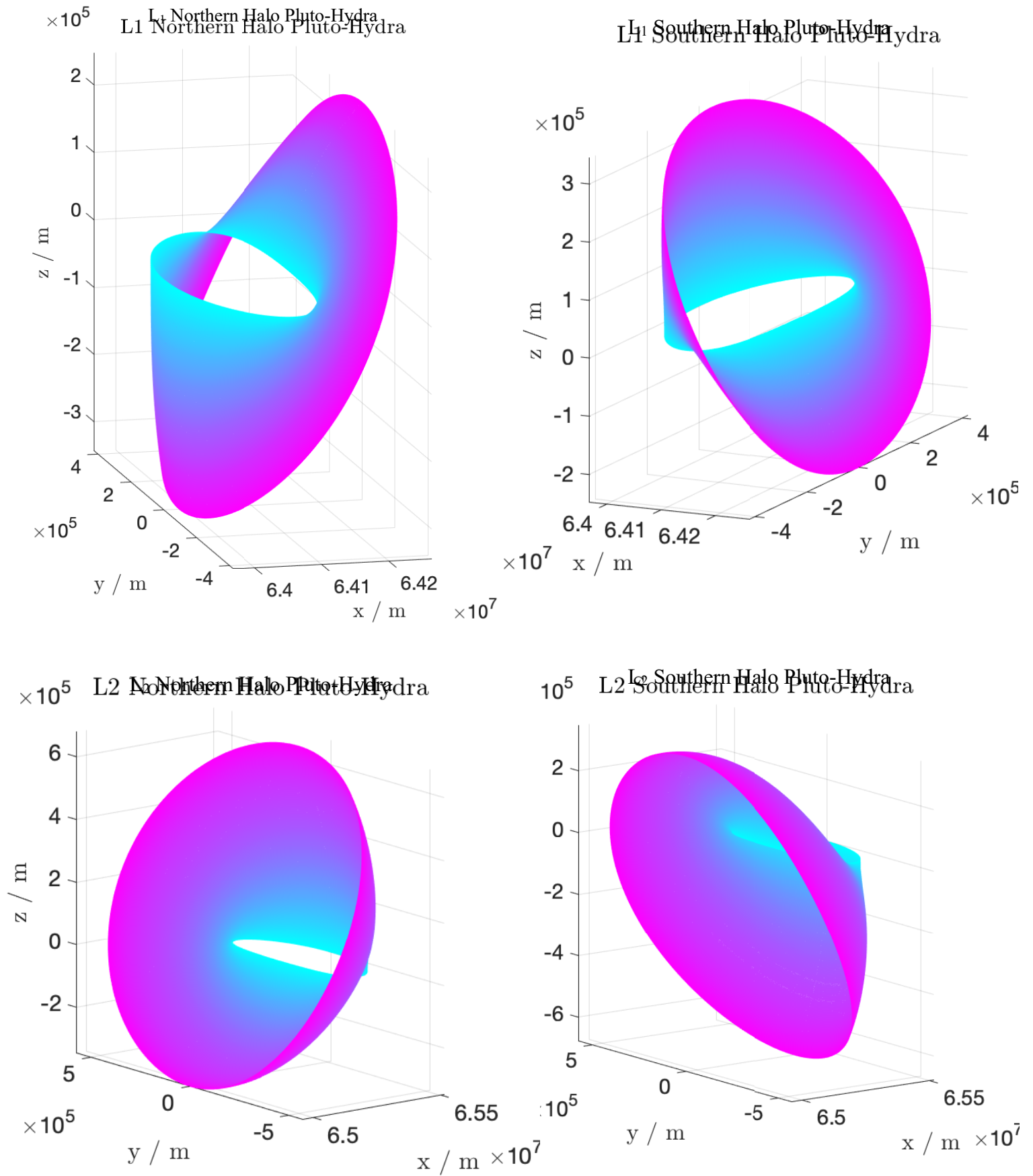


Figure 28: Northern and Southern Halo Families for Pluto-Hydra

Pluto-Kerberos – $\mu = 0.00000122793$

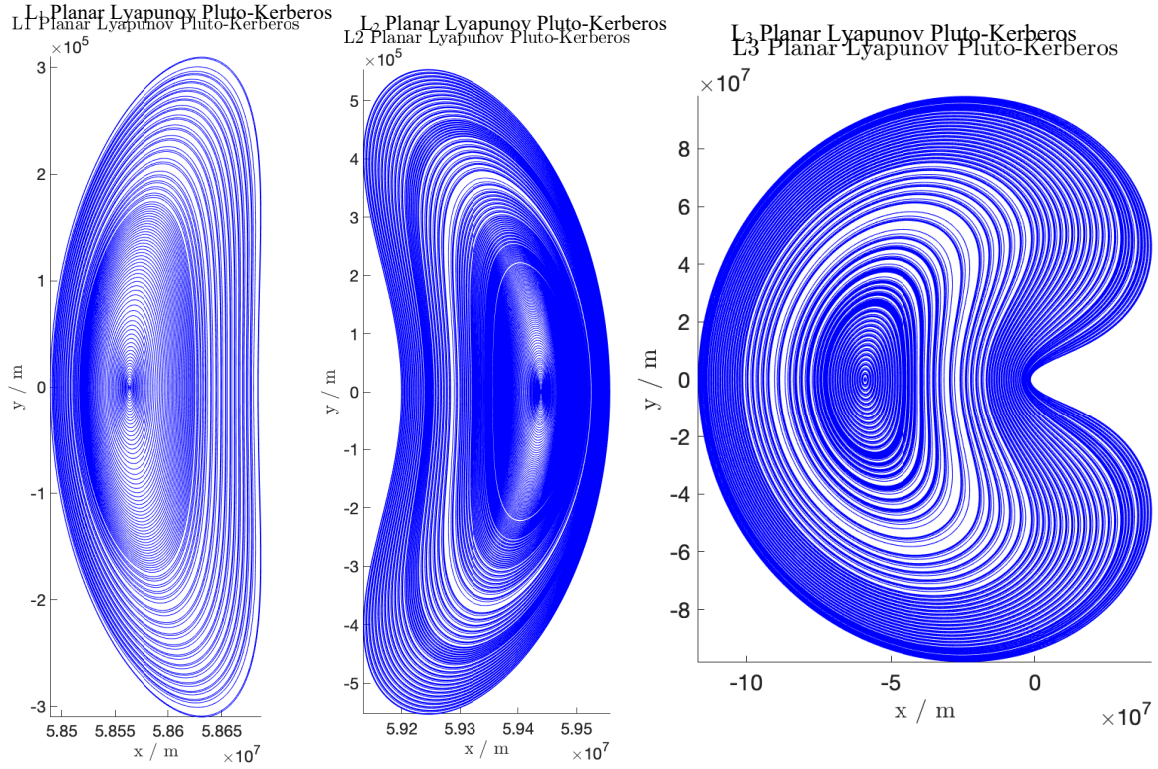


Figure 29: Planar Lyapunov family for Pluto-Kerberos

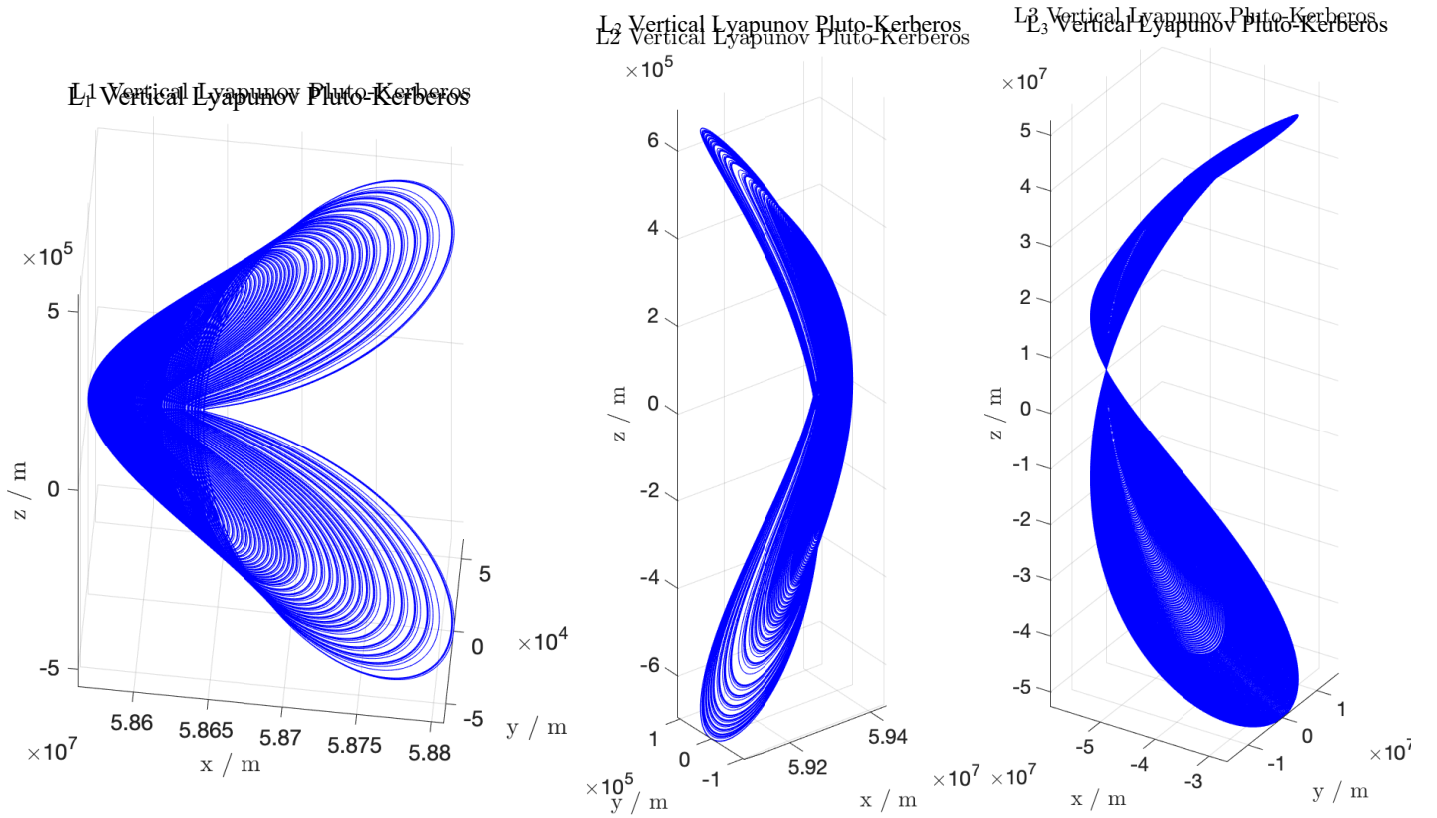


Figure 30: Vertical Lyapunov family for Pluto-Kerberos

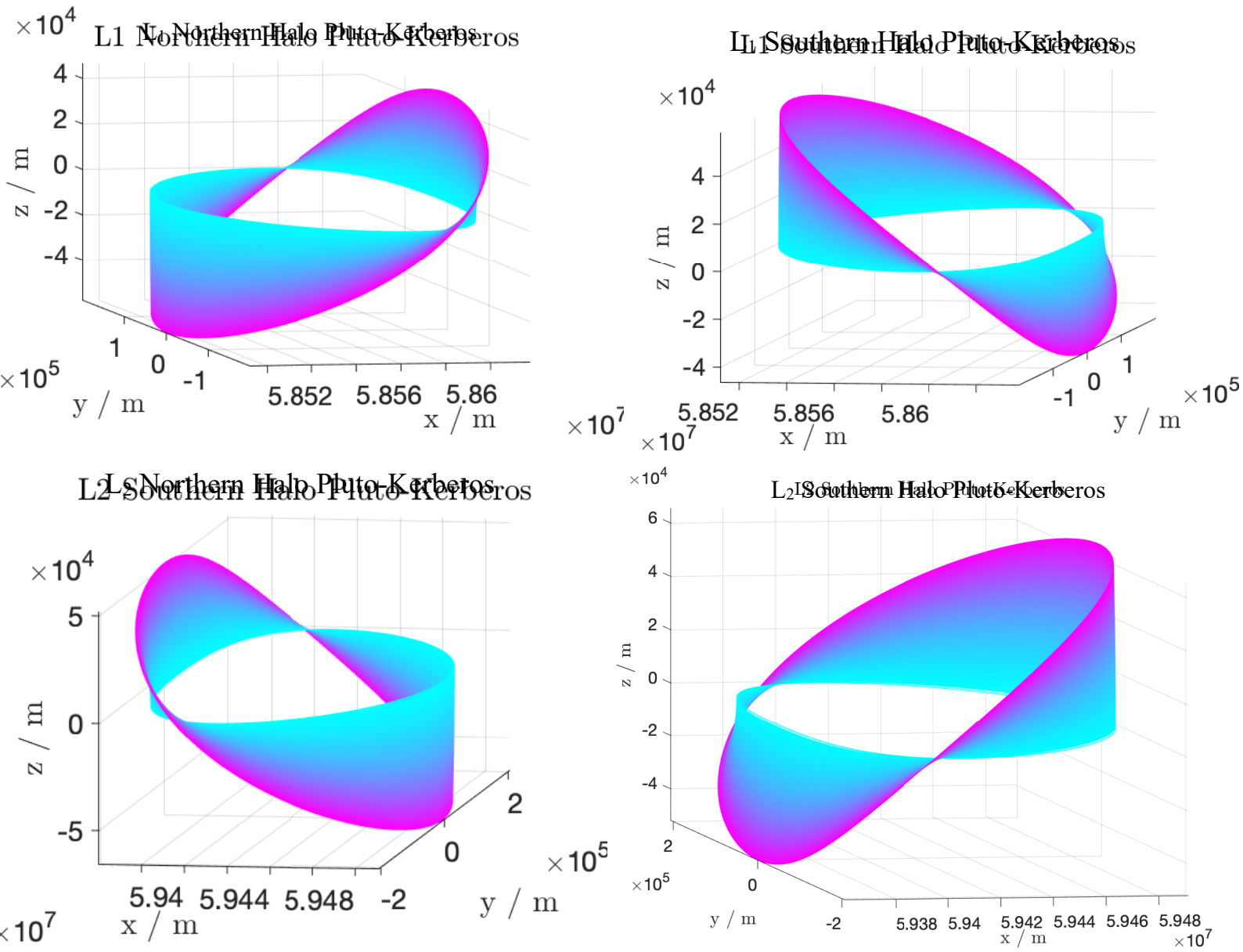


Figure 31: Northern and Southern Halo Families for Pluto-Kerberos

Eris Dysnomia – $\mu = 0.00488618758$

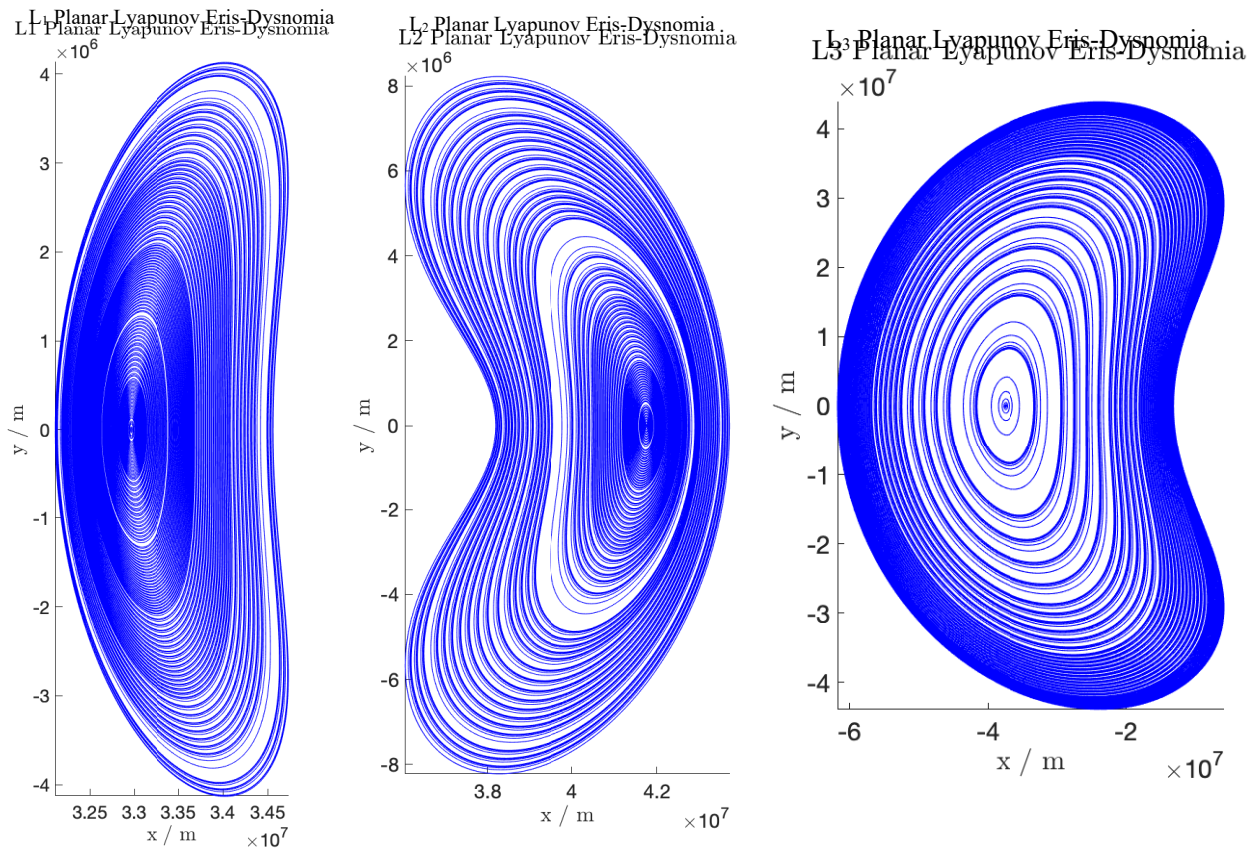


Figure 32: Planar Lyapunov family for Eris-Dysnomia

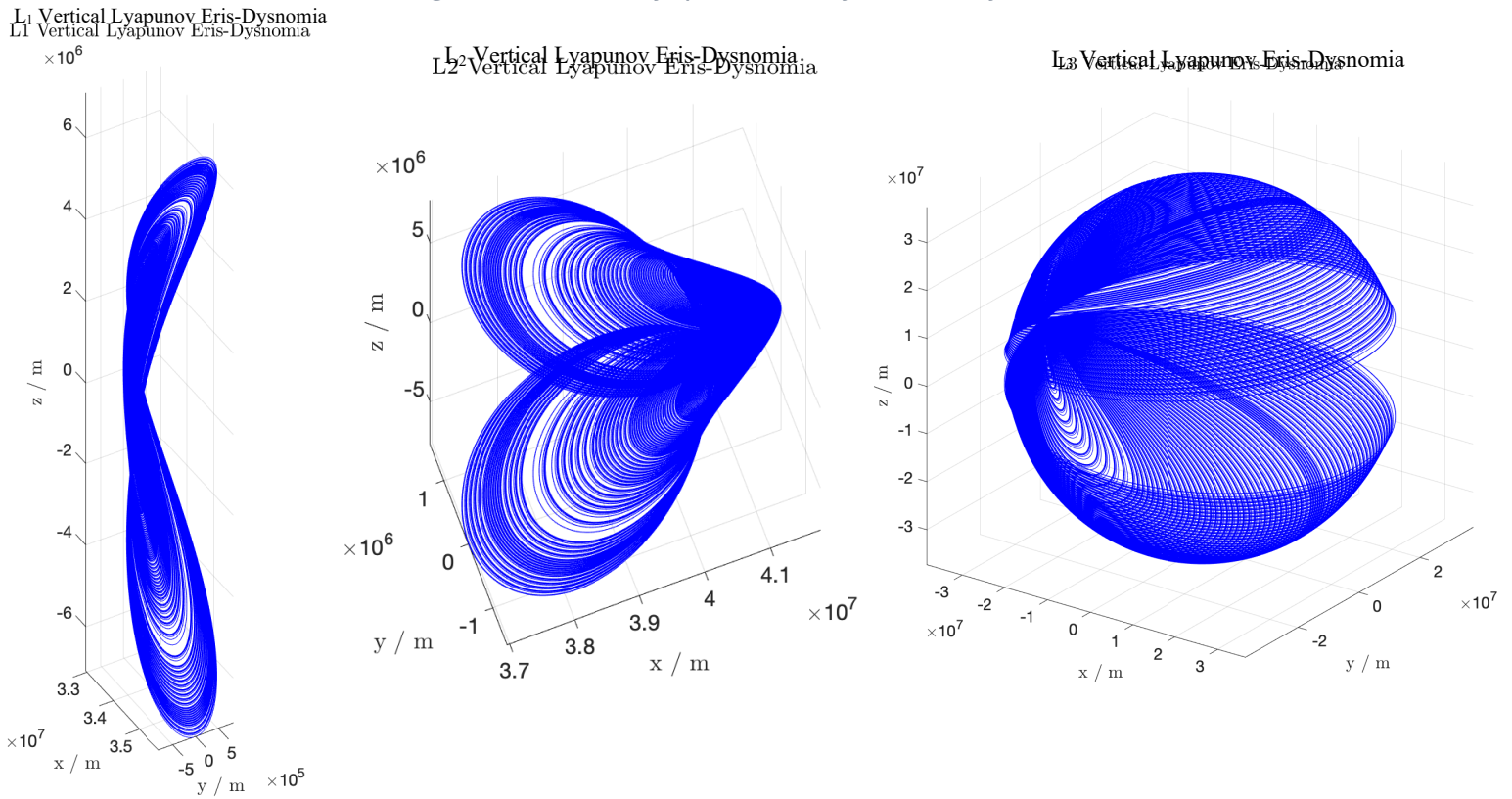
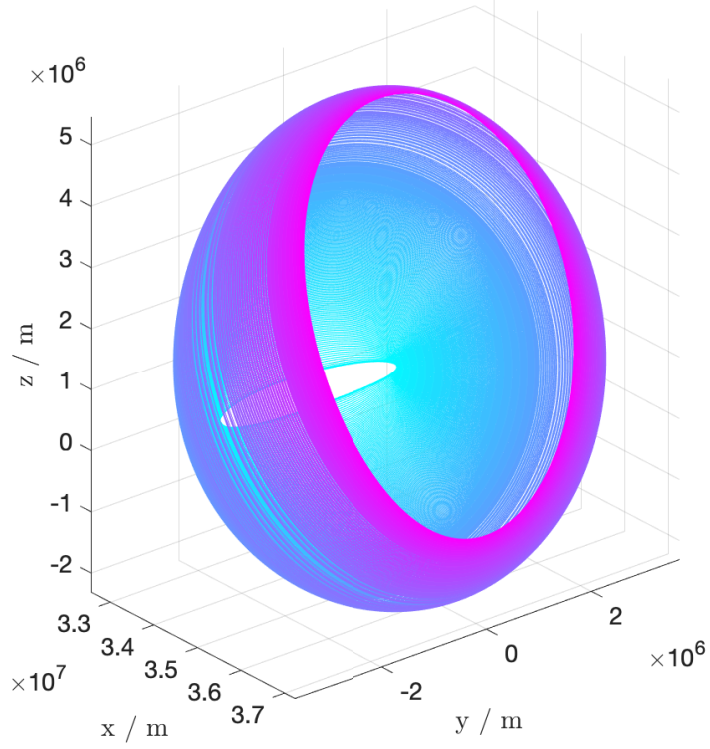
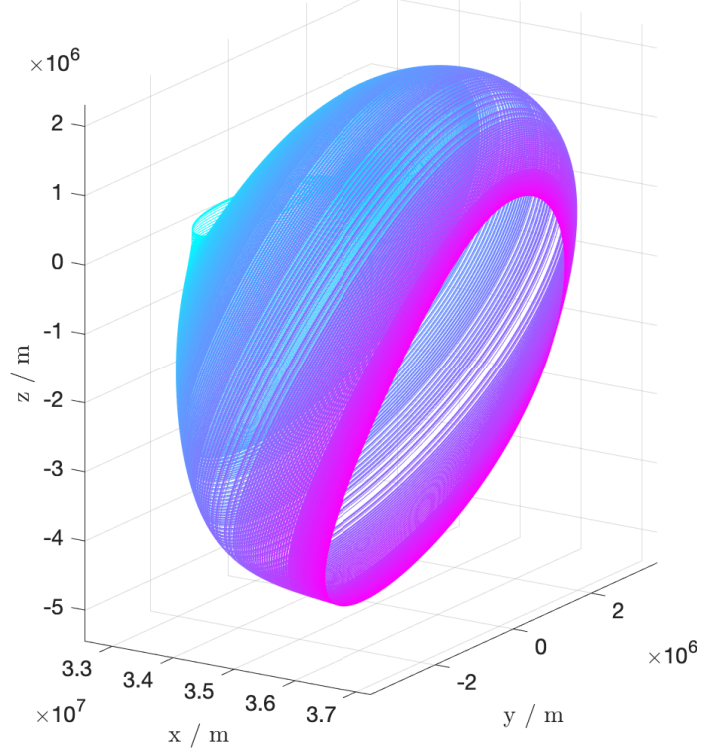


Figure 33: Vertical Lyapunov family for Eris-Dysnomia

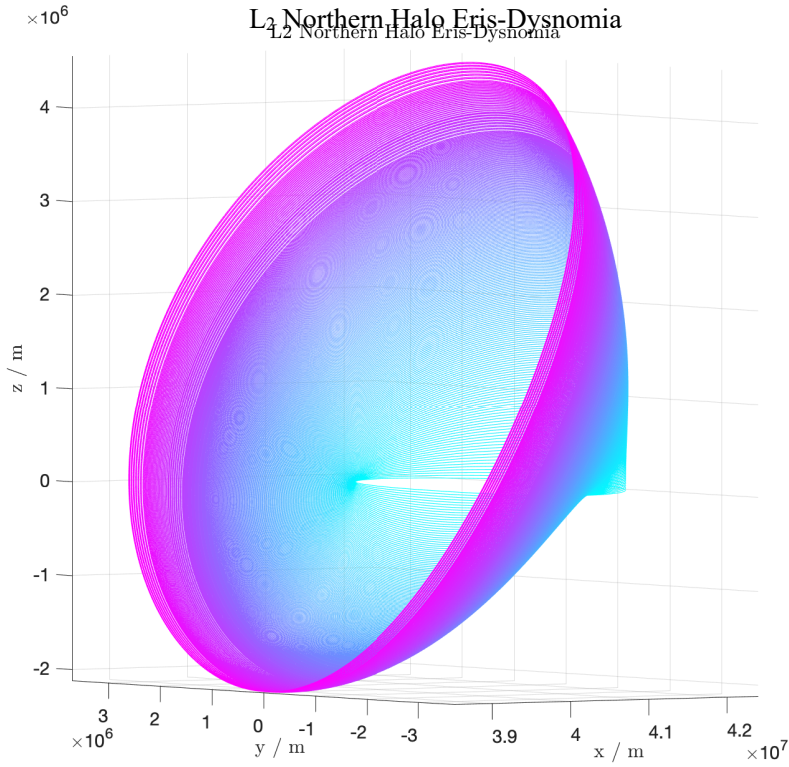
L₁ Northern Halo Eris-Dysnomia



L₁ Southern Halo Eris-Dysnomia



L₂ Northern Halo Eris-Dysnomia



L₂ Southern Halo Eris-Dysnomia

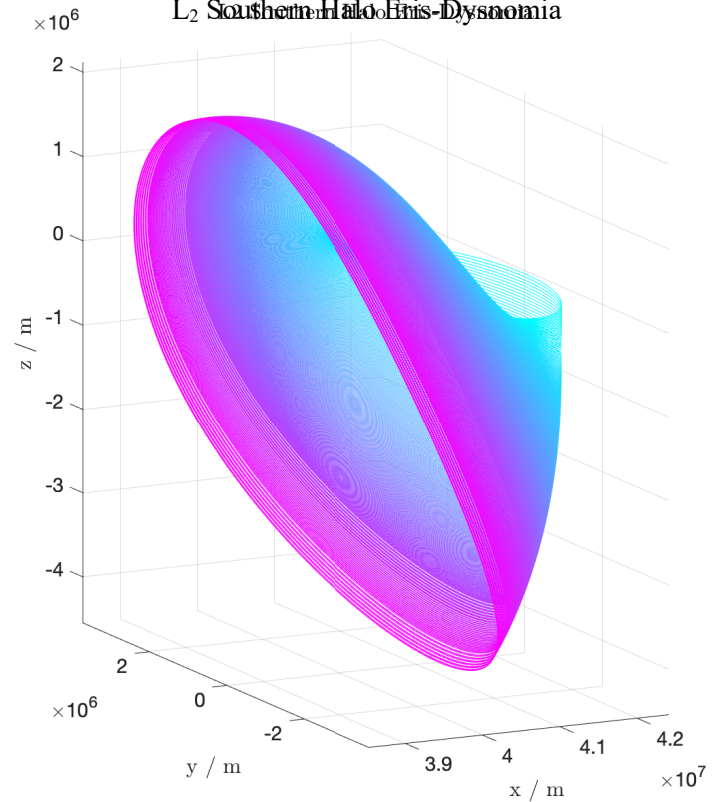
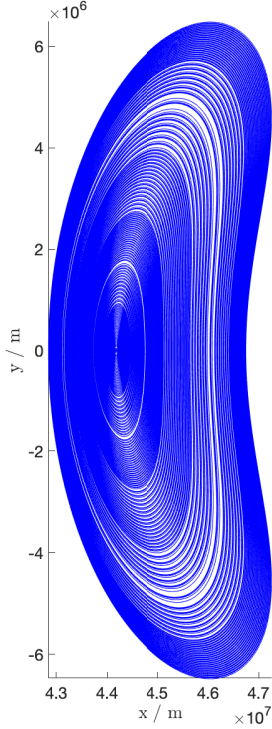


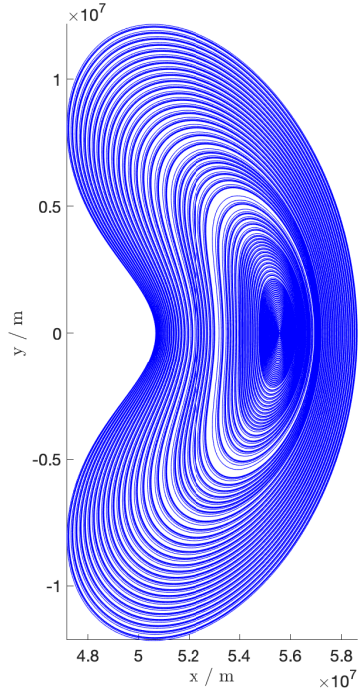
Figure 34: Northern and Southern Halo Families for Eris-Dysnomia

Haumea-Hi'iaka– $\mu = 0.00447984071$

L1 Planar Lyapunov Haumea-Hi'iaka



L2 Planar Lyapunov Haumea-Hi'iaka



L3 Planar Lyapunov Haumea-Hi'iaka

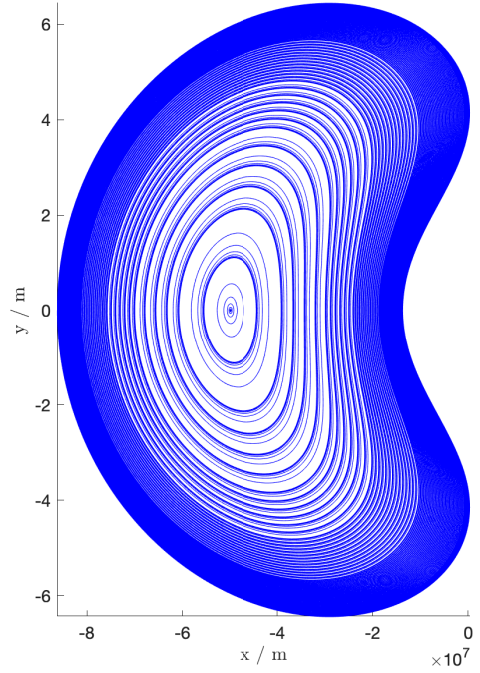
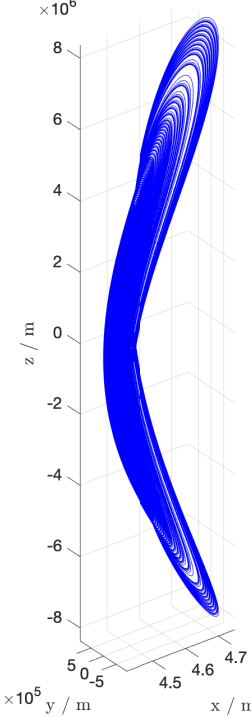
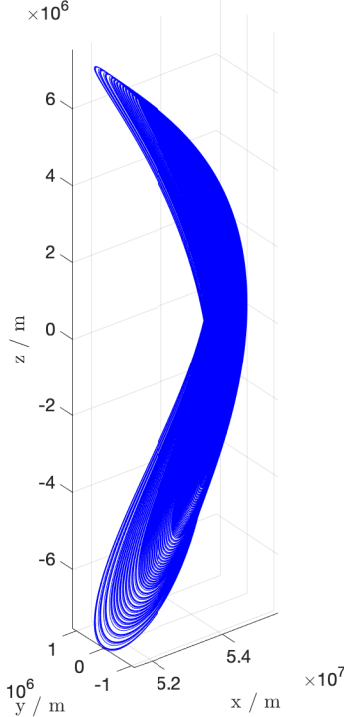


Figure 35: Planar Lyapunov family for Haumea-Hi'iaka

L1 Vertical Lyapunov Haumea-Hi'iaka



L2 Vertical Lyapunov Haumea-Hi'iaka



L3 Vertical Lyapunov Haumea-Hi'iaka

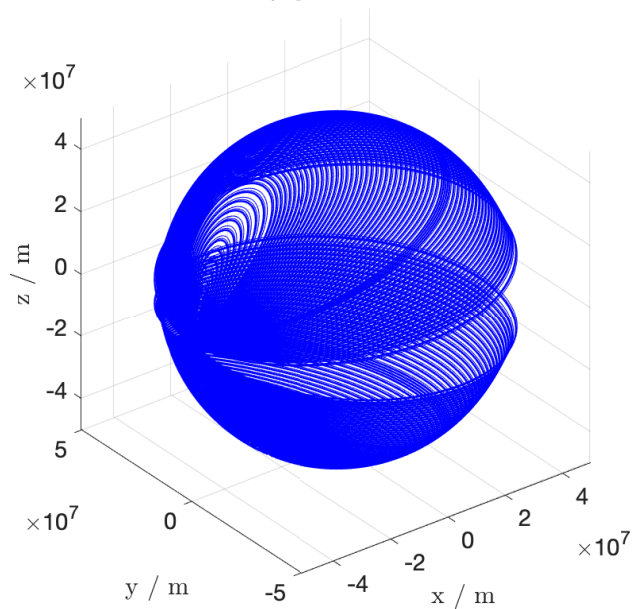
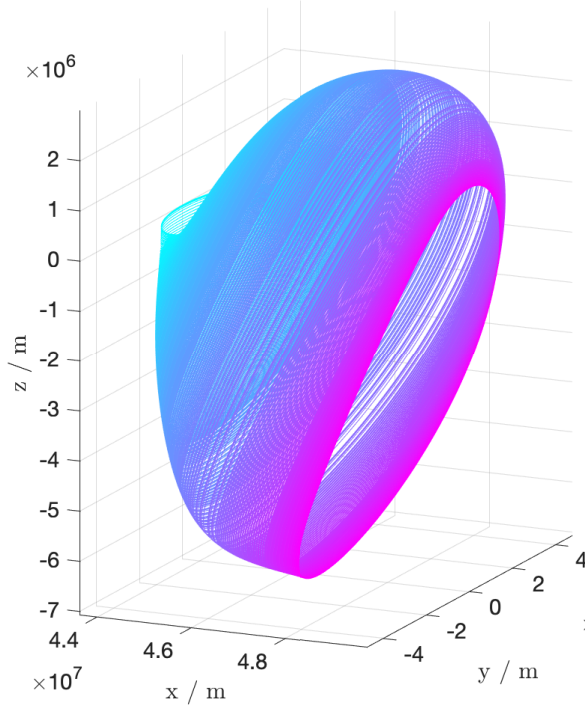
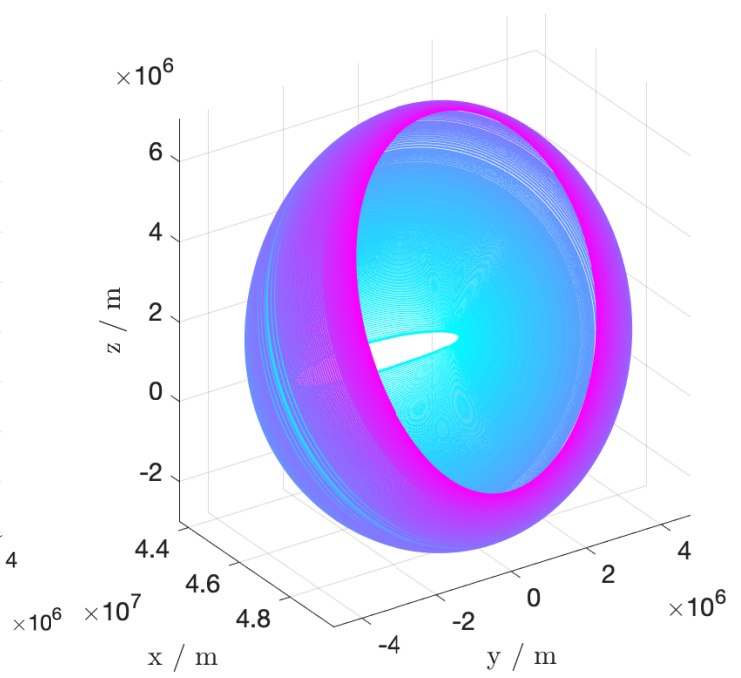


Figure 36: Vertical Lyapunov family for Haumea-Hi'iaka

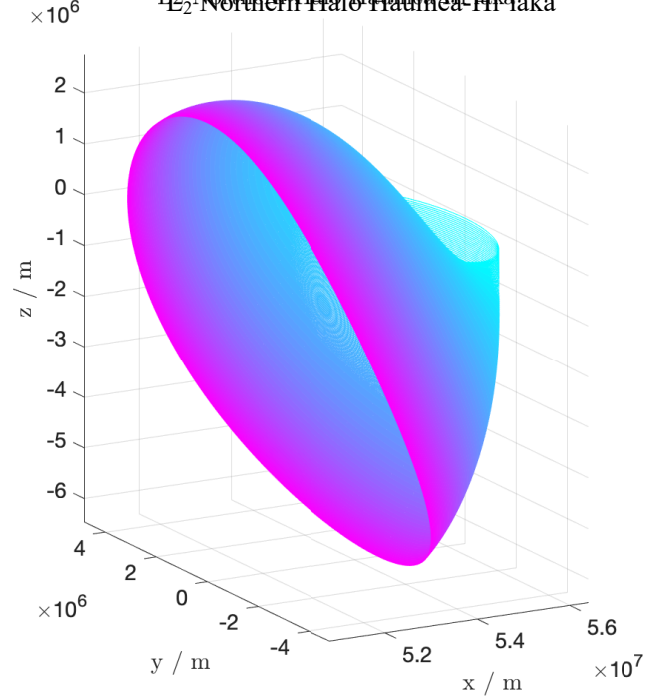
L₁ Northern Halo Haumea-Hi'iaka



L₁ Southern Halo Haumea-Hi'iaka



L₂ Northern Halo Haumea-Hi'iaka



L₂ Southern Halo Haumea-Hi'iaka

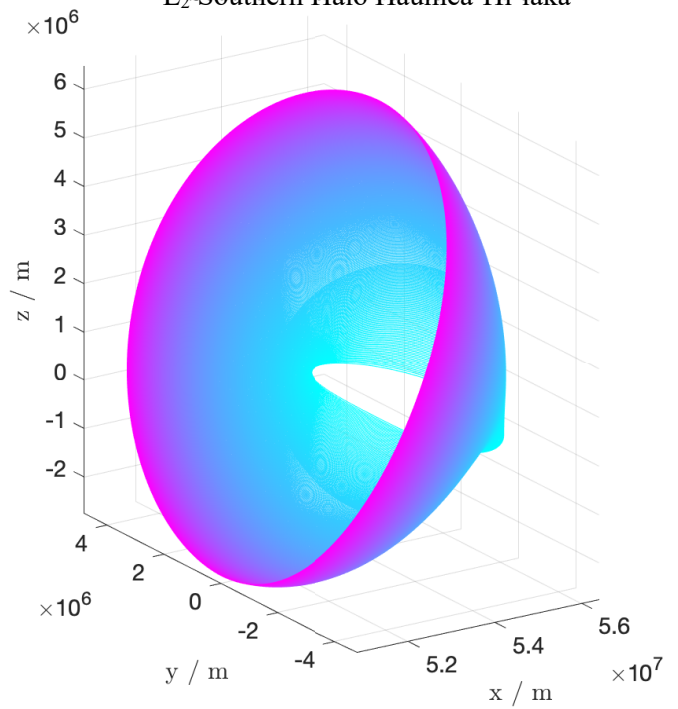


Figure 37: Northern and Southern Halo Families for Haumea-Hi'iaka

Haumea-Namaka – $\mu = 0.00049975012$

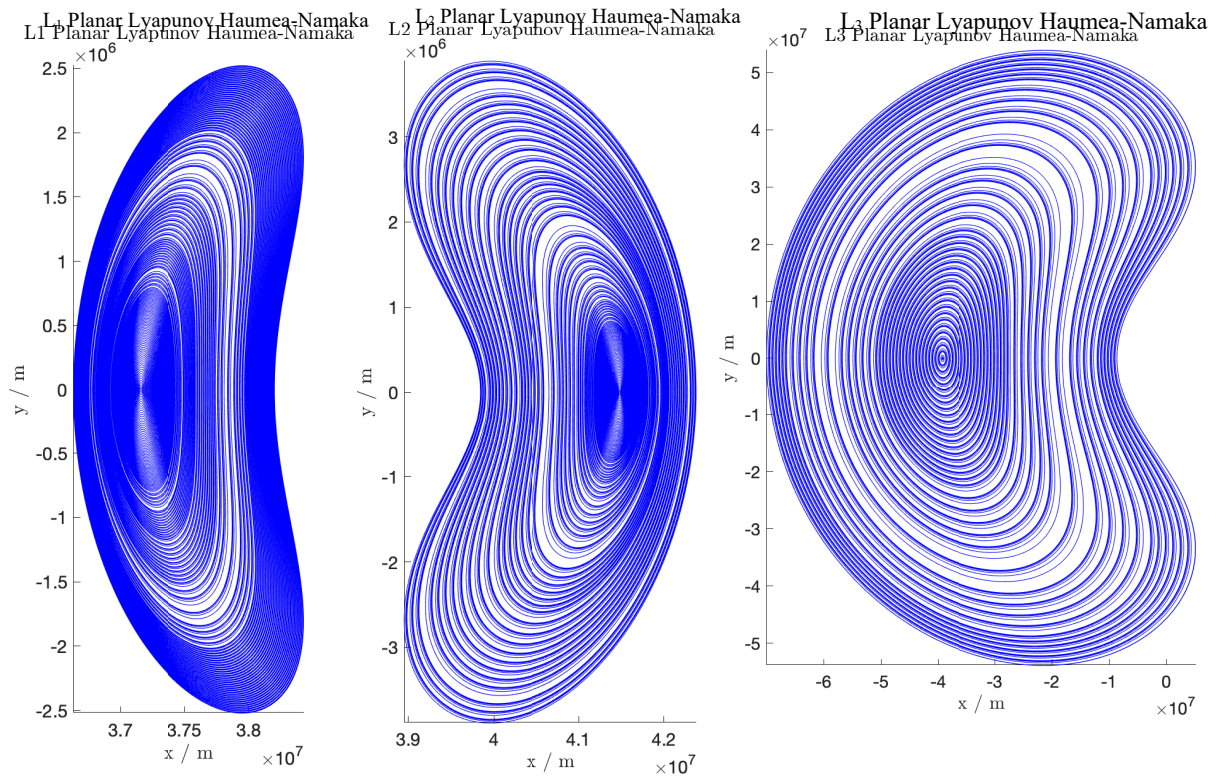


Figure 38: Planar Lyapunov family for Haumea-Namaka

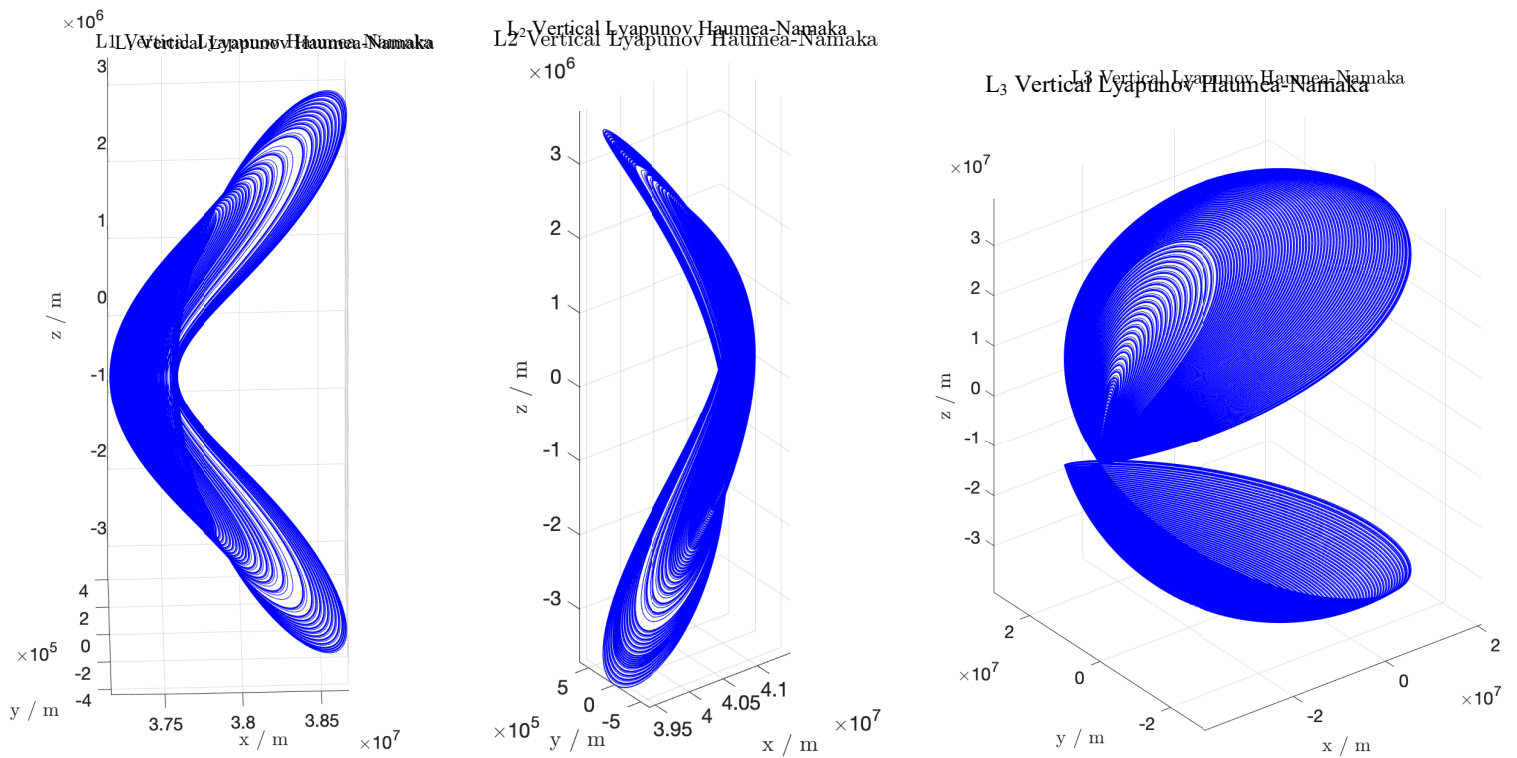


Figure 39: Vertical Lyapunov family for Haumea-Namaka

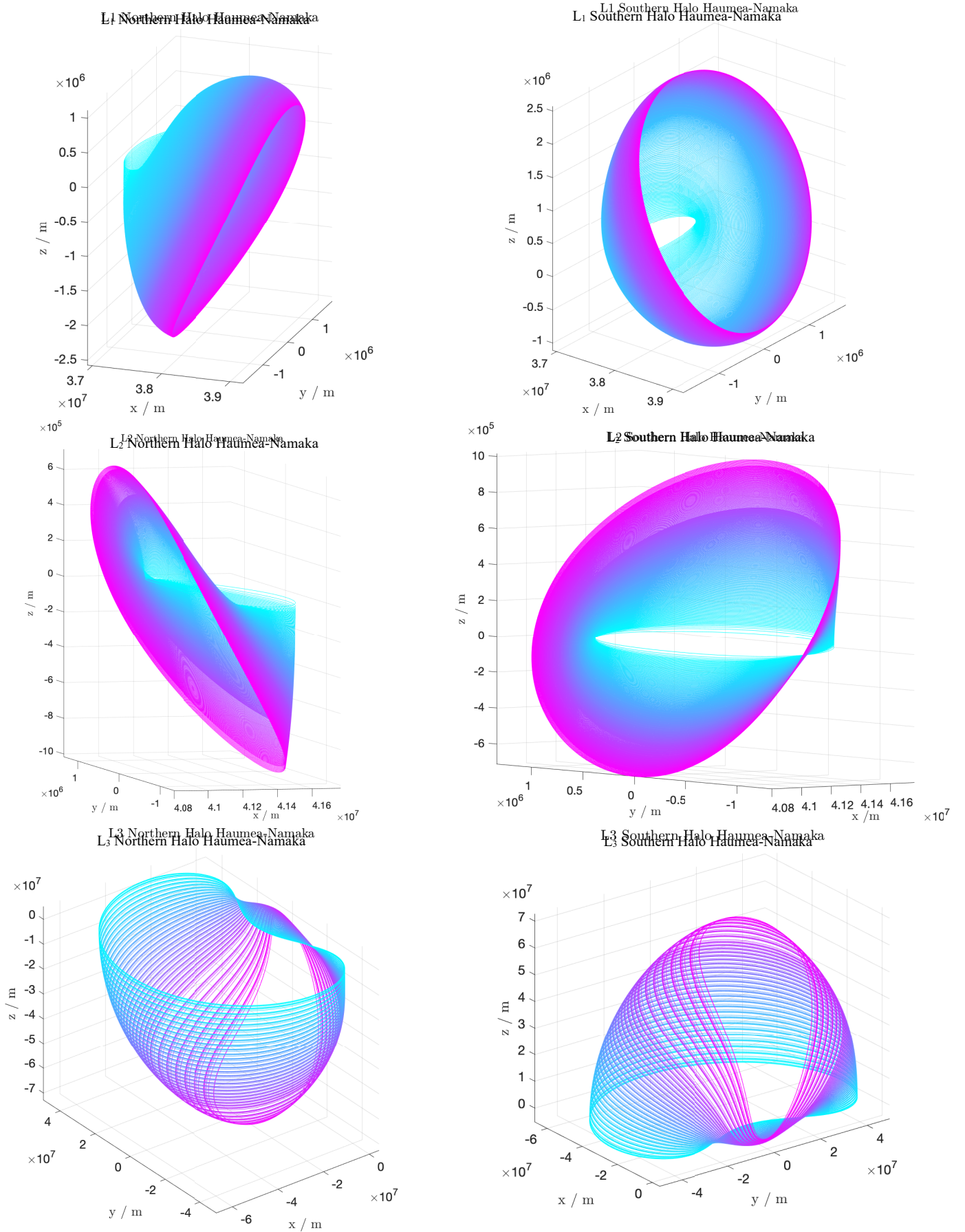


Figure 40: Northern and Southern Halo Families for Haumea-Namaka

APPENDIX II – Flowcharts for Other Processes

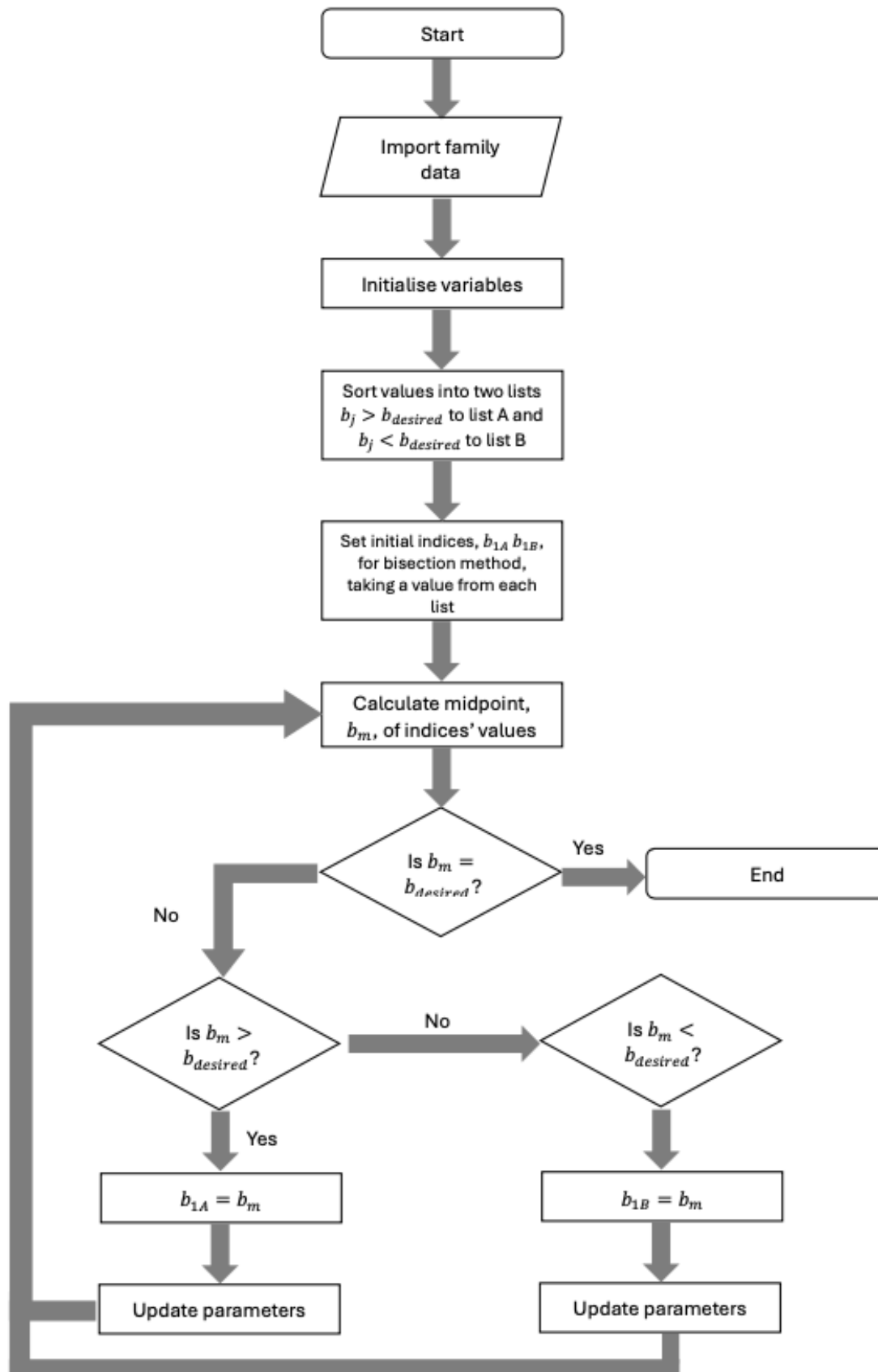


Figure 41: Flowchart representing the bisection method discussed in section 3.7.1